Who this course is for?

Intended audience and pre-requisites for this course

WE'LL COVER THE FOLLOWING

- Intended Audience and Pre-requisites

First things first: thanks for having chosen this course. We hope reading it will be both beneficial and pleasurable to you.

Intended Audience and Pre-requisites

This course is targeting developers that already have a basic understanding of what React is and how it works. It's not meant to be used as a complete how-to guide but as an introduction to popular concepts/design patterns, so it's suggested that you take an introductory level course in React if you have not already.

The course is a representation of one out of many understandings and interpretations of the described patterns, so it essentially articulates one approach to patterns in React.

The next chapter will give a brief overview of what you should expect to learn in this course

Learning outcomes

A brief summary of what you would learn in this course

At the end of this course, you will learn about:

- The different ways of passing props and getting output
- Composition and higher-order components
- Presentational and container components
- Describing the concept and importance of one-way direction data flow
- Implementing one-way direction data flow using Flux and Redux
- Properly managing dependencies in your React applications
- Style React components
- Integrating third-party libraries in you React applications

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Let’s dive right into it!

Foundational Elements of React

Before we dive into the patterns of React, let's revise the building blocks of a React component.

WE'LL COVER THE FOLLOWING

- JSX
- The `Render()` function
- Props

JSX

JSX is nothing more than syntactic sugar on top of JavaScript, allowing you to write what seems to be HTML but in JavaScript. Have a look at the following code snippet

```
import React from 'react';
require('./style.css');

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

Lines 6 to 10 use HTML code that gets transpiled into JavaScipt and executed.

The `Render()` function

The `render` function is mandatory in all React components. This function is invoked by React in order to generate output, that is, function calls that end up modifying the DOM. Inside `render`, `this` is referencing the component instance (ex: `this.props`, `this.state`)

> **REMEMBER:** `render` must always return something, `JSX` tags or `false/null` in cases where the component should not output any content. JSX syntax must always be enclosed in a single root node. In order to avoid the unnecessary creation of divs or other elements, React introduced a special component called `Fragment` that can wrap multiple nodes without creating an actual DOM element around them.

Props

Props represent the input values of a component. They are passed as attributes on `JSX` and are accessed through `this.props`. We'll talk about them in more detail in the next chapter.

```
import React from 'react';

export default class Greeting extends React.Component {
    render() {
        return <span>Hello {this.props.name}!</span>;
    }
}
```

Any valid JavaScript element can be sent as a prop: string, boolean, number, object, function, array, etc. Here we send the `name`, a string on **line** 7 as a prop.

> **REMINDER:** Props should never be *modified* by the component!

Now that we have revised a basic component of React i.e. components (pun fully intended), let's move on to how `state` is used in React.

State in React

If props represent outside data for a component, its state represents the internal data.

WE'LL COVER THE FOLLOWING

- What is `state`
- How `setState()` works
- Quick quiz on `state`

What is `state`

Each component can encapsulate a `state` object which is accessible inside all the functions but is not exposed to the outside world. Similar to `props`, you can add primitives as well as complex structures to a `state` object.

How `setState()` works

The state should not be mutated directly but through the `setState()` function!

In order to change any value inside the `state` object, we do the following call:

```
this.setState({key: newValue});
```

With this, the `key` is either updated in the state or created if it didn't exist before. All other keys are preserved, so you don't need to specify them each time you call `setState`.

`setState` also accepts a function as a parameter:

```
this.setState((oldState) => ({ value: oldState.value + 1 }));
```

In this example, we are increasing the `value` with one, but we need the old value in order to compute the new one. Even if you're not using the `oldState`, this is the recommended way of working with `setState`.

this is the recommended way of working with `setState`.

A component that handles its state would look like this:

```
import React from 'react';

export default class Message extends React.Component {
    constructor() {
        super();
        this.state = {
            showMessage: true,
        };

        this.toggle = this.toggle.bind(this);
    }

    toggle() {
        this.setState(state => ({
            showMessage: !state.showMessage,
        }));
    }

    render() {
        return (
            <div>
                {this.state.showMessage && "Message is visible"}
                <button onClick={this.toggle}>Click me!</button>
            </div>
        );
    }
}
```

The constructor creates the state with the key `showMessage` and value `true` on **lines 6-8**. Then the `toggle()` function inverts the value of `showMessage` by setting the state on **lines 14-16**. The `render` function creates a div on **lines 22-24** that displays a message if `showMessage` is `true`. Curly braces inside JSX, as on **line 22**, indicate that the code inside them is to be treated as JavaScript code. So, the string "Message is visible" is only displayed if `this.state.showMessage` is `true`. Then, a button is created on **line 23** that calls the `toggle()` function when clicked which causes the message to disappear/reappear.

> **REMINDER**: `setState` does not perform an instant action and the state object will not be updated on the next line. Also, behind the curtains, `React` tries to optimize the number of re-renders by batching up multiple calls to `setState`.

Quick quiz on `state`

1 Which of the following is correct?

COMPLETED 0%

1 of 5

Now that we have revised some foundational elements and state in React, let's study lifecycle hooks in the next lesson.

The Component Lifecycle

These functions allow us to tap into the lifecycle of a component and add custom code at given times.

WE'LL COVER THE FOLLOWING

- Lifecycle Methods
- Quick quiz on the component lifecycle

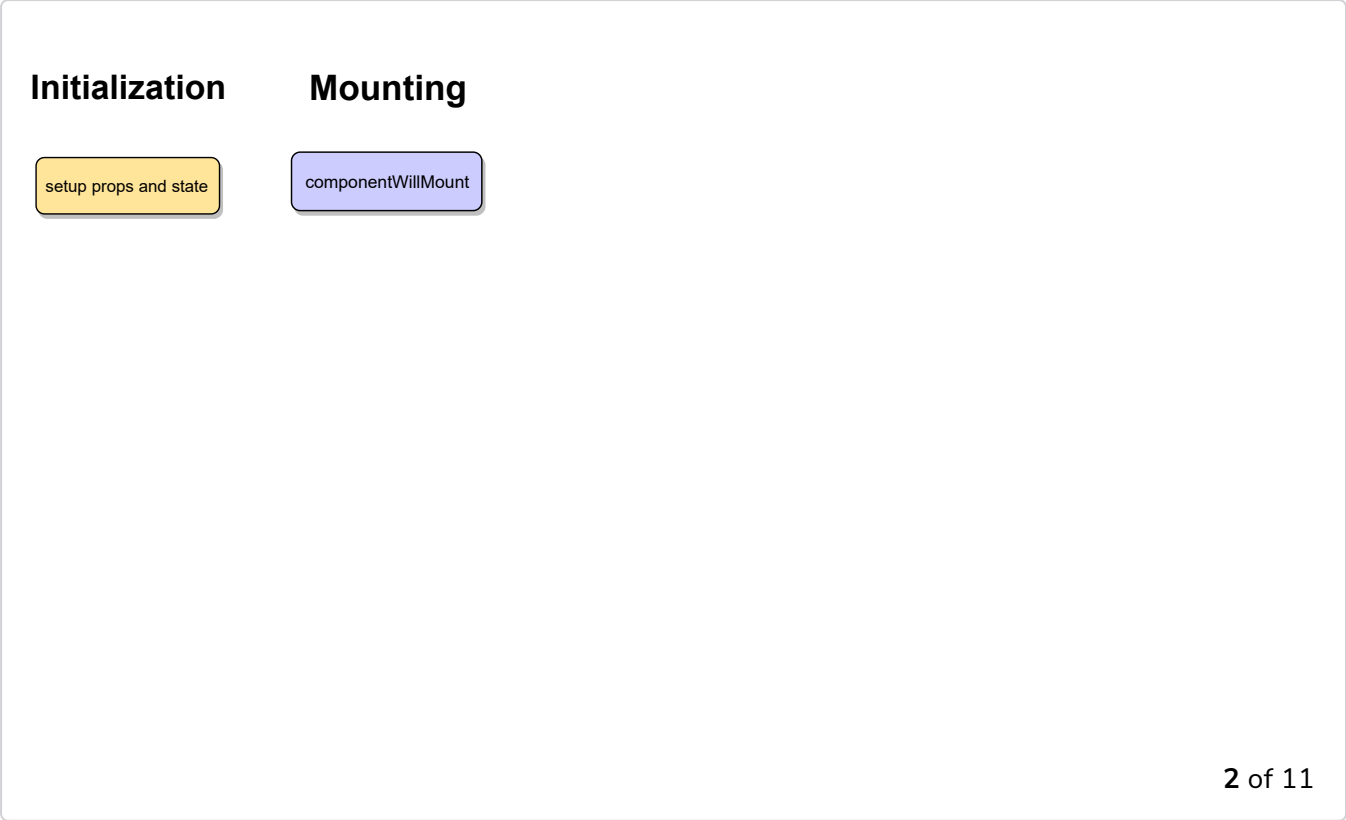
Lifecycle Methods

For each stage in the life of a component, we have specific functions that we can optionally implement in our components.

Function	When its called
`constructor`	called just once when the component is instantiated.
`componentDidMount`	called just once, after the component is added for the first time to the DOM, used for adding event handlers or making asynchronous data calls.
`componentWillUnmount`	called just once when the component is due to be removed from the DOM, used for cleanup.
`shouldComponentUpdate`	method used for component-level performance optimization

`componentWillReceiveProps` (deprecated)	replaced by `getDerivedStateFromProps`, called before the mount and also after each component re-render.

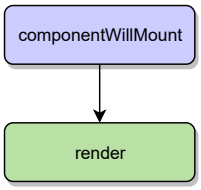
The animation below will show you when each hook is being called by React:



Initialization

setup props and state

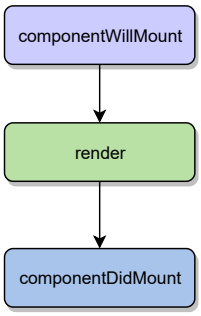
Mounting

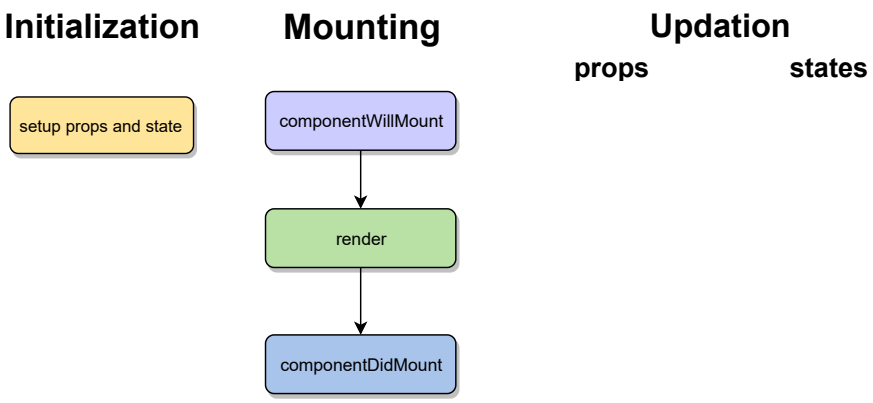


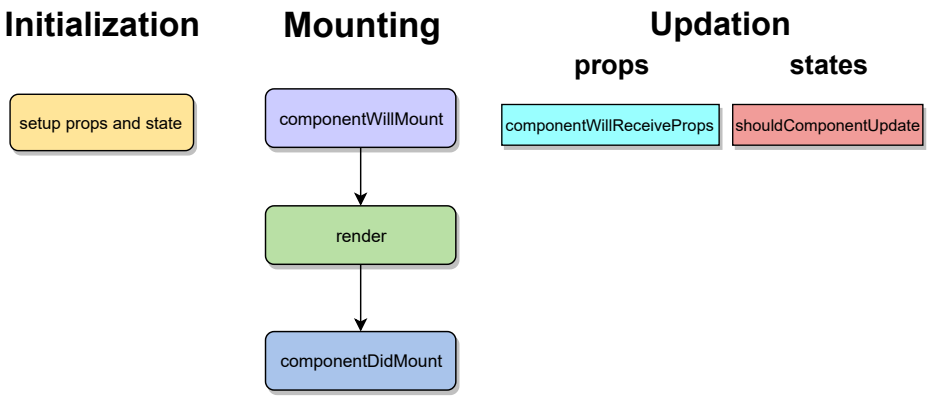
Initialization

setup props and state

Mounting




Initialization
Mounting
Updation
props
states
setup props and state
componentWillMount
render
componentDidMount


Initialization
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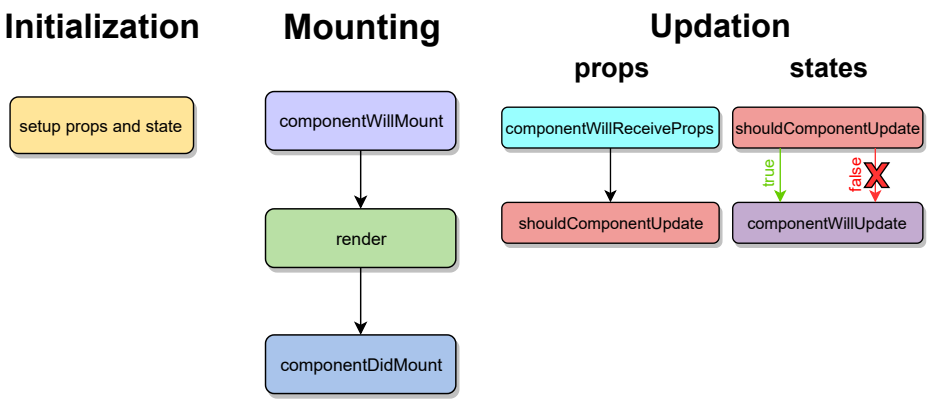
setup props and state
componentWillMount
render
componentDidMount
componentWillReceiveProps
shouldComponentUpdate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componentDidMount
componentWillReceiveProps
shouldComponentUpdate
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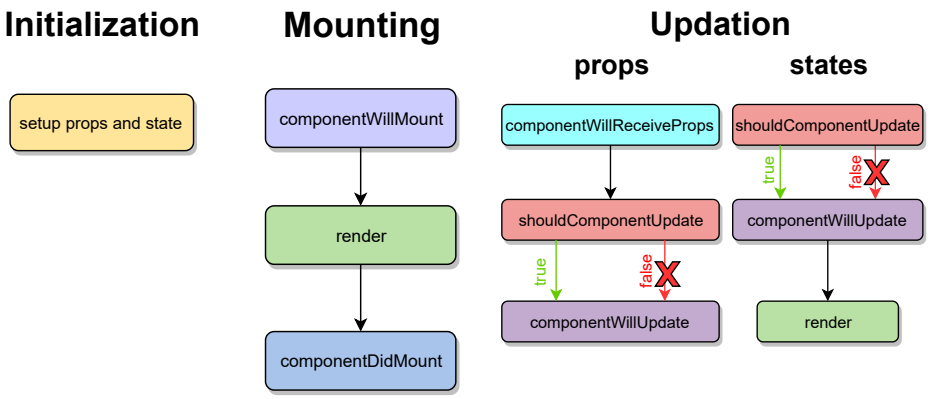
true
false
componentWillUpdate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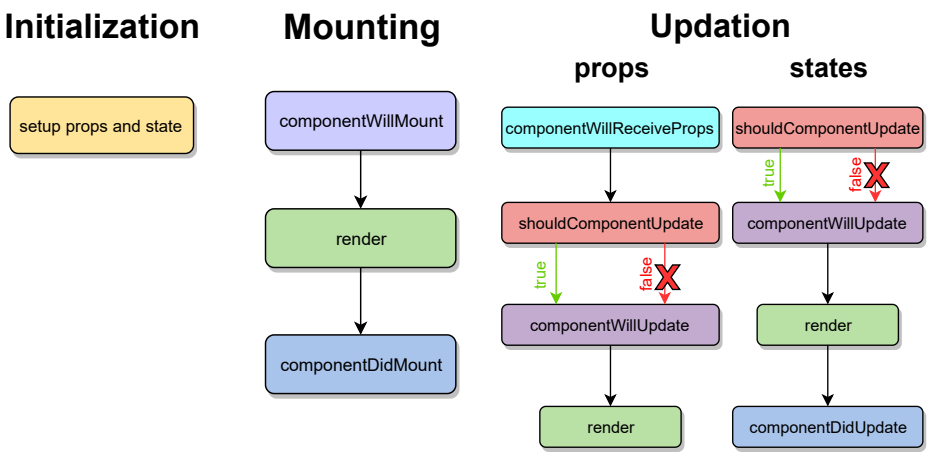
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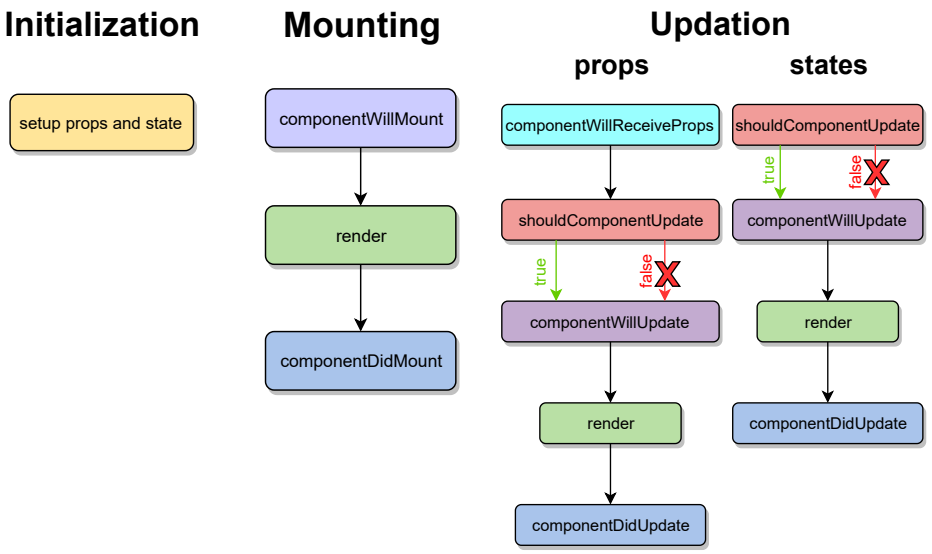
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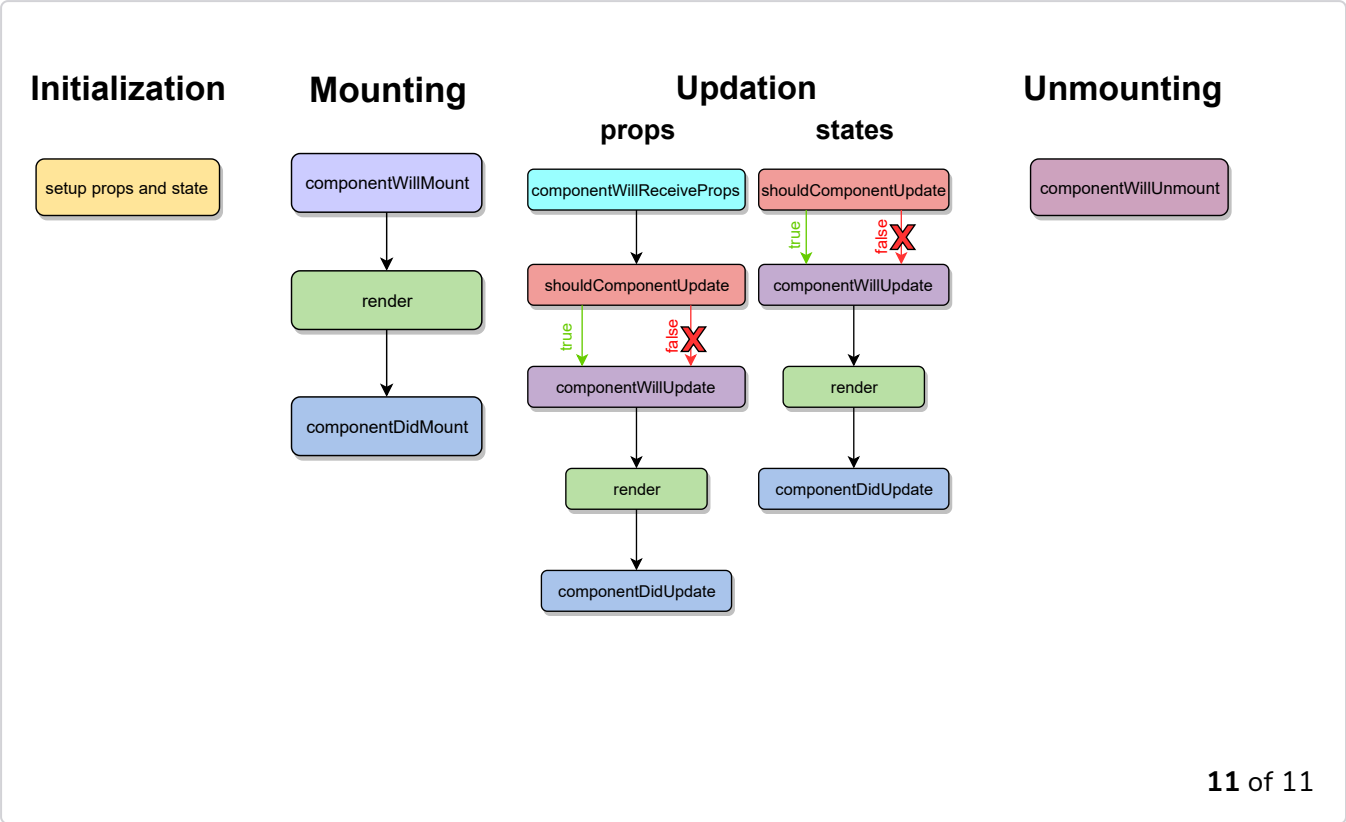
componentWillUpdate
render
componentDidUpdate



You can run the code below and check the **console** to see the messages from the lifecycle methods.

```
import React from "react";

class Greeting extends React.Component {
    constructor() {
        super();
        console.log("created");
    }

    componentDidMount() {
        console.log("mounted");
    }

    componentWillUnmount() {
        console.log("unmounted");
    }

    static getDerivedStateFromProps(newProps) {
        console.log("props received", newProps);
    }

    render() {
        return (
            <div>
                <span>Hello {this.props.name}!</span>
            </div>
        );
```

```
    }
}


export default Greeting;
```

Quick quiz on the component lifecycle

1 Which of the following is not a phase of React's component lifecycle?

COMPLETED 0%

1 of 3

Now that we understand the lifecycle of react components let's move onto how communication works in React.

More on Props

An introduction to react components and how they communicate with one another.

WE'LL COVER THE FOLLOWING

- `propTypes`
- Passing components as props
- Quick Quiz on Props!

Every React component is like a small system that operates on its own. It has its own state, input, and output. In this section, we will explore these characteristics.

React Component

As discussed in the previous chapter, the input of React components are **props**, which are the means through which data is passed to them. For example, in the following code playground, the `Title` component has only one input or prop - `text`. The parent component should provide it as an attribute while using the `<Title>` tag. The parent component in the example below is `App` and it passes "Hello React" to it.

```
import React from 'react';
import Title from './title';

export default class App extends React.Component {
  render() {
    return (
      <div>
          <Title text="Hello React" />
      </div>
```

```
      );
  }
}
```

`propTypes`

In addition to the component definition, we also have to define, at the very least, `propTypes`. Within `propTypes`, we define the type of every property and React provides hints in the console if something unexpected gets sent. Have a look at **lines 19-22** in `titles.js` where the `text` prop is set to be a string.

`defaultProps` is another useful option. We may use it to set a default value of a component's props so that if the developer forgets to pass them, we have meaningful values available. Have a look at **lines 24-27** to see how to use `defaultProps`. Try removing the input passed to `Title` in the `App` component and running the code!

Passing components as props

React does not strictly define what should and shouldn't be passed as a prop. It may be anything we want. It could even be another component,

```
import React from 'react';

export default class SomethingElse extends React.Component {
  constructor(props) {
    super(props);
  }

  render() {
  return <div>The answer is { this.props.answer }</div>;
  }
}
```

Here, `Answer` is passed to `SomethingElse` as a prop on **line 9** in `index.js`.

There is also a `props.children` property that gives us access to the child components passed by the owner of the component. For example:

```
import React from 'react';
import PropTypes from 'prop-types';

export default class Title extends React.Component
{

  render()
  {
    return (
      <h1>
```

```
        {this.props.text}
        {this.props.children}
      </h1>

    );
  }
};
Title.propTypes =
{
  text: PropTypes.string,
  children: PropTypes.any

};

Title.defaultProps =
{
  text: PropTypes.string
};
```

In this example, `<span>community</span>` in the `App` component is `children` in the `Title` component. Notice that if we don't return `{ children }` as part of the `Title`'s body the `<span>` tag will not be rendered.

(prior v16.3) An indirect input to a component may also be the so-called `context`. The whole React tree may have a `context` object which is accessible by every component. We will further elaborate on that in the dependency injection section.

Quick Quiz on Props!

1 You can pass functions, components, and text as props

COMPLETED 0%

1 of 3

Output from React Components

In this lesson, we'll be looking at how to get output from React components.

WE'LL COVER THE FOLLOWING

- Output
- Using lifecycle methods to trigger a process

Output

The first obvious output of a React component is the rendered HTML. Visually, that is what we get. However, because the prop may be anything, including a function, we could also send out data or trigger a process.

In the following example, we have a component that accepts the user's input and sends it out (`<NameField />`).

```
import React from 'react';
import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

Using lifecycle methods to trigger a process

Very often we need an entry point of our logic. React comes with some handy lifecycle methods that may be used to trigger a process. For example, say we have an external resource that needs to be fetched on a specific page.

```
import React from 'react';
import Results from 'results';

const API = 'https://hn.algolia.com/api/v1/search?query=';
const DEFAULT_QUERY = 'redux';
```

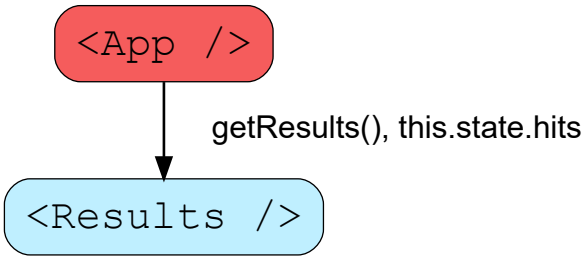
```
const DEFAULT_QUERY = 'redux';
export default class App extends React.Component {
  constructor() {
    super();
    this.getresults = this.getresults.bind(this);
    this.state = {
      hits: [],
    }
  }
  getresults()
  {
    fetch('https://hn.algolia.com/api/v1/search?query=redux')
      .then(response => response.json())
      .then((json) => {
        this.setState({ hits: json.hits });
     });
  };
render() {
    return <Results hits={this.state.hits} getResults={this.getresults}/>
  }
}
```

The code above renders a loading screen while data is fetched from an API. The `App` component passes the function `getresults()` as a prop to the `Results` component. `App`'s state's `hits` attribute is also passed as a prop to `Results`. `getresults()` is called within the `Results` component when it *mounts* by calling it within the lifecycle method `ComponentDidMount()` on **lines 8-10**. The `getresults()` function updates `App`'s state and when it updates, `App` rerenders thereby updating the props passed to `Results`. This causes the condition within the if-statement to become true and hence rendering the fetched JSON data. Have a look at the slides below for a better understanding.

Another variation of this could be building a search-results experience. We have a search page and we enter our criteria there. We click submit and the user goes to `/results` where we have to display the result of the search. Once we land on the results page we render some sort of a loading screen and trigger a request for fetching the results in `componentDidMount` lifecycle hook. When the data comes back we pass it to a `<List>` component.

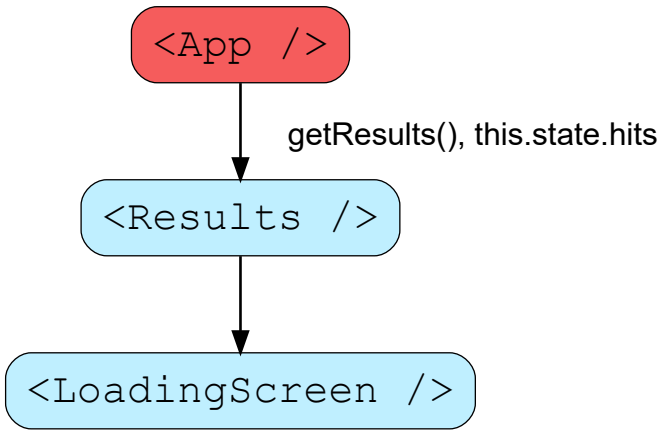
```
<App />
```

Output

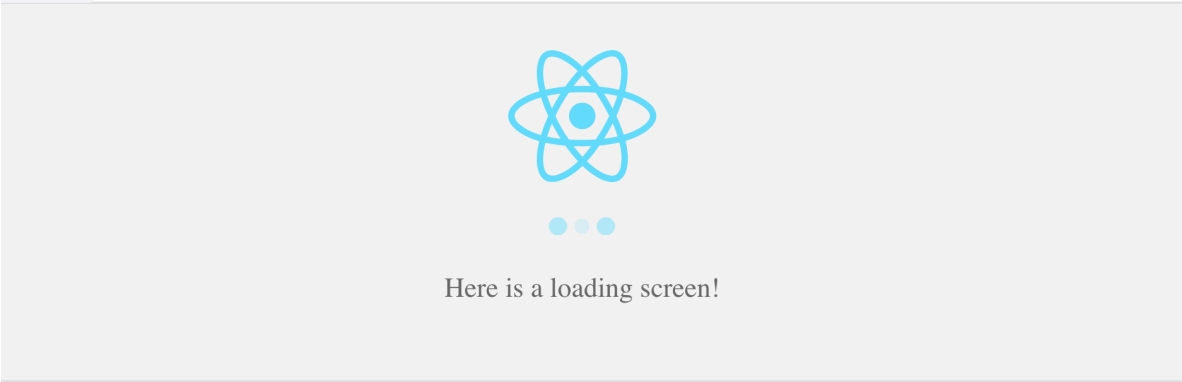


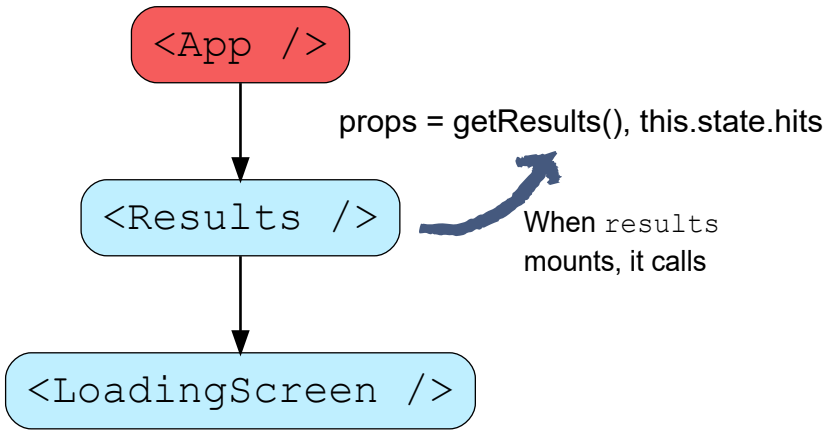
Output

Generating output...

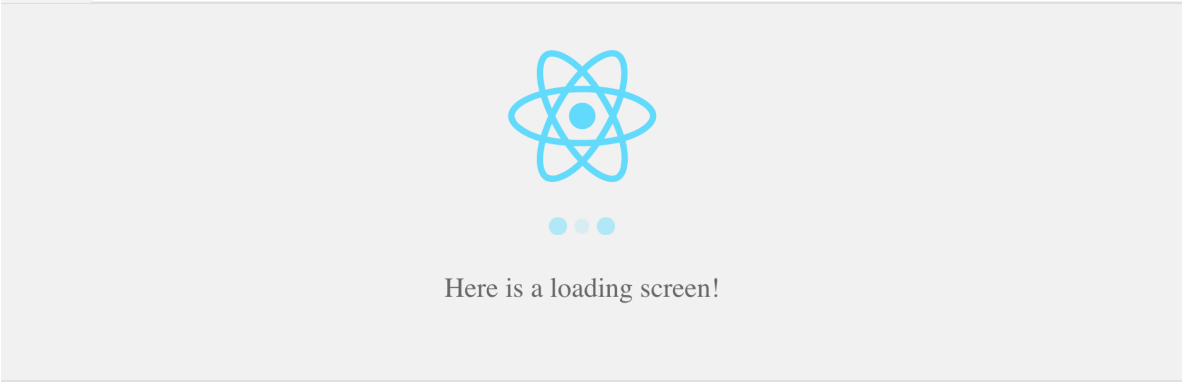

<App />
getResults(), this.state.hits
<Results />
<LoadingScreen />

Output


Here is a loading screen!


<App />
props = getResults(), this.state.hits
<Results />
When results mounts, it calls
<LoadingScreen />

Output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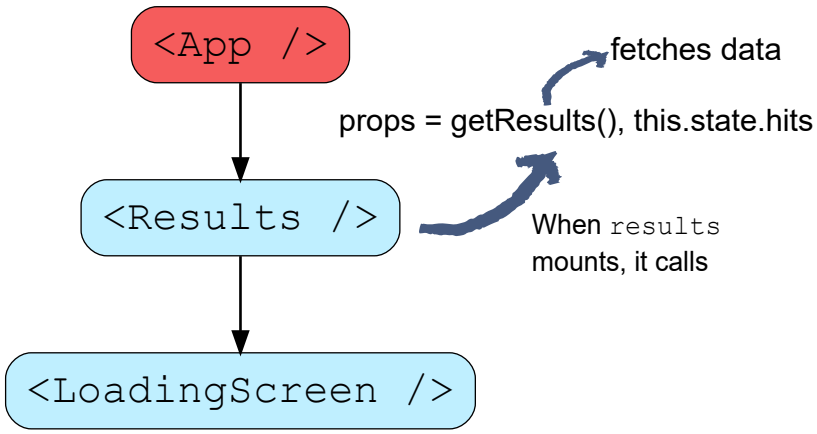
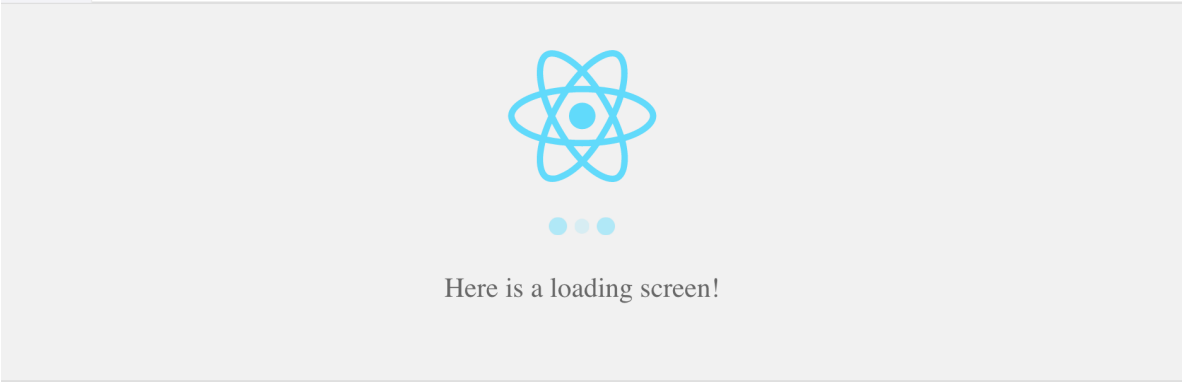

<App />
fetches data
props = getResults(), this.state.hits
<Results />
When results mounts, it calls
<LoadingScreen />

Output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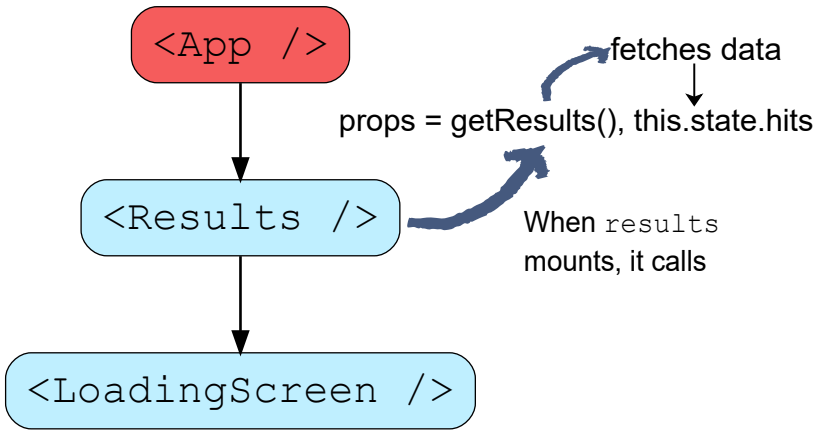
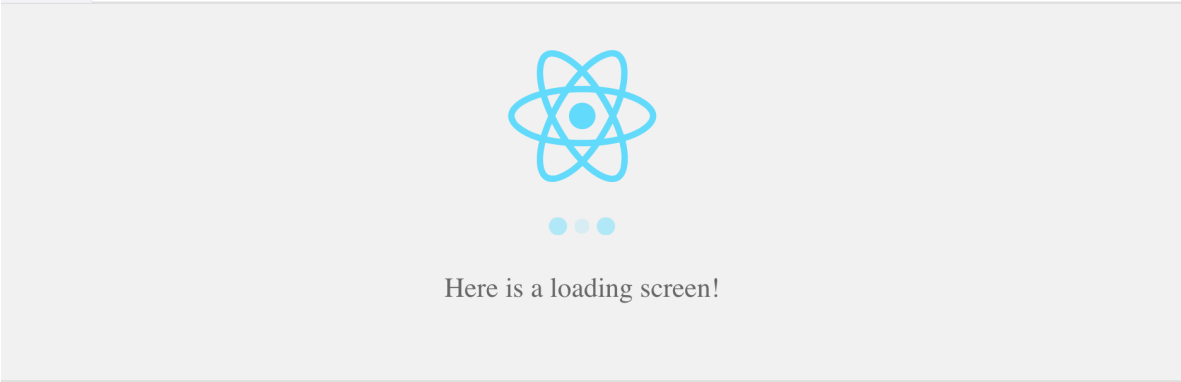
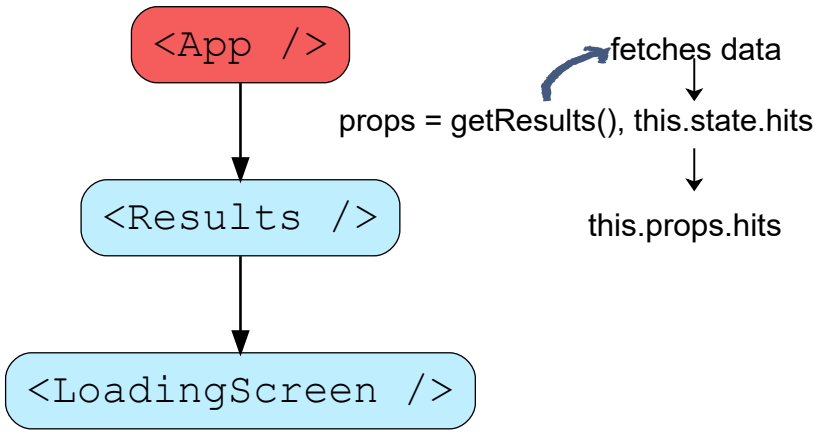
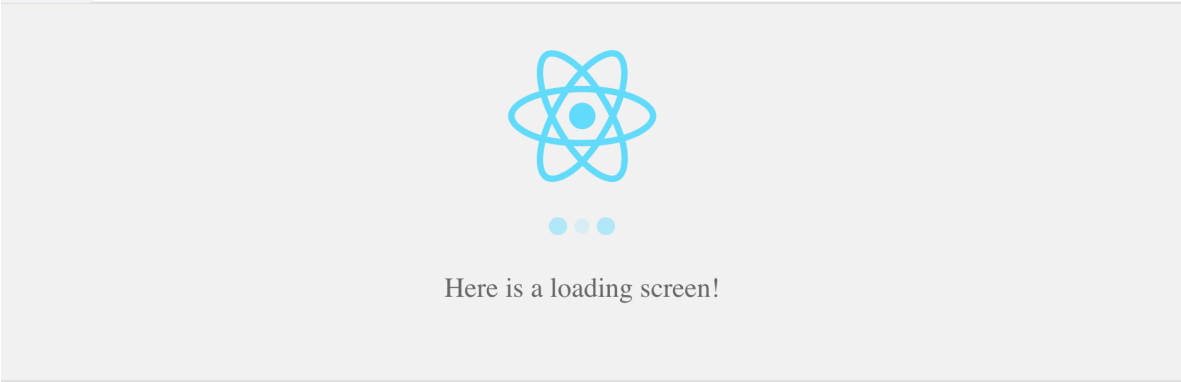

<App />
fetches data
props = getResults(), this.state.hits
<Results />
When results mounts, it calls
<LoadingScreen />

Output


Here is a loading screen!


<App />
fetches data
props = getResults(), this.state.hits
<Results />
this.props.hits
<LoadingScreen />

Output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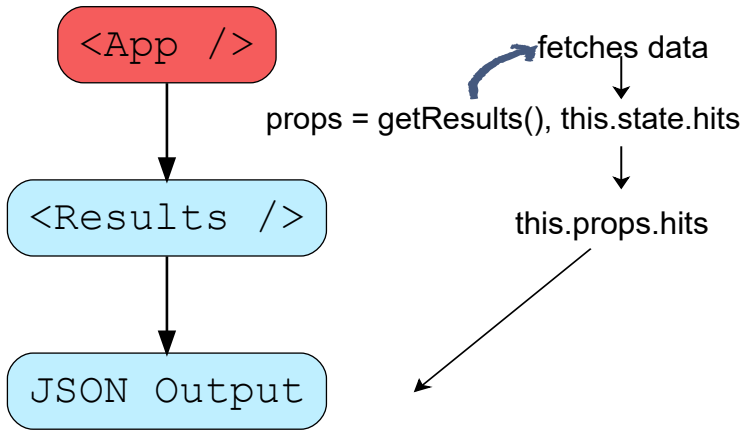


Output

ld Yourself a Redux

ɔs://zapier.com/engineering/how-to-build-redux/

ngs to learn in React before using Redux

ɔs://www.robinwieruch.de/learn-react-before-using-redux/

ɔw HN: ORY Editor – A rich editor for the browser, built with React and Redux

ɔs://github.com/ory/editor?branch=master

ɔundCloud client in React and Redux

ɔ://www.robinwieruch.de/the-soundcloud-client-in-react-redux/

Now that we know how to get output from React components, lets do a few exercises in the next lesson!

Test Your Skills So Far!

Test yourself on React!

WE'LL COVER THE FOLLOWING

- Final thoughts
- Coding Challenge: Test your React Skills
- Quick Quiz on Getting React Output!

Final thoughts

It is nice that we may think about every React component as a black box. It has its own input, lifecycle, and output. It is up to us to compose these boxes. And maybe that is one of the advantages that React offers: easy abstraction and composition

For instance, the code below is the “black box” that we have defined as the Title component to take in multiple child components as inputs and output each of them:

```
import React from 'react';
import PropTypes from 'prop-types';

class Title extends React.Component {
  constructor(props) {
    super(props);
    this._clickHandler = this._clickHandler.bind(this);
  }
  _clickHandler() {
    console.log(this.props.children);
  }
  render() {
    return (
      <h1>
        <a onClick={this.props.logoClicked}>
          <img src='
        </a>
        { this.props.text }
        { this.props.children }
```

```
        <a onClick={this._clickHandler}>click on the logo</a>
      </h1>
    );

  }
};
Title.propTypes = {
  logoClicked: PropTypes.func,
  text: PropTypes.string,
  children: PropTypes.any
};

class SomethingElse extends React.Component {
  render() {
    return <div>The answer is { this.props.answer }</div>;
  }
}
SomethingElse.propTypes = {
  answer: PropTypes.element
};

class Answer extends React.Component {
  render() {
    return <span>42</span>;
  }
}

export default class NextApp extends React.Component
{
  logoClicked()
   {
    console.log('logo clicked');
  }
  render() {
    return (
      <Title text='Hello React ' logoClicked={this.logoClicked}>
        <span>community</span>
        <SomethingElse answer={ <Answer /> } />
      </Title>
    );
  }
};
```

Coding Challenge: Test your React Skills

Since this is the first coding challenge of this course, lets start off with something relatively easy. Write a react component called ‘App’ that returns a span with the text ‘Hello World’. Some skeleton code is already given.

```
import React from 'react';
import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

Quick Quiz on Getting React Output!

1 Components cannot refer to other components in their output.

COMPLETED 0%

1 of 4

Now that we have covered basic communication in React components, we will move on to handling events in React components.

Event Handlers

How event handlers in React differ from the standard DOM.

WE'LL COVER THE FOLLOWING

- Comparison With Standard DOM
- Using The Bind Function
- Final thoughts
- Quick quiz on Event Handlers!

In the previous section, we covered React communication in React components. Building upon that, we will now discuss how events are handled in React.

Comparison With Standard DOM

React provides a series of attributes for handling events. The process is very similar to the one used in standard DOM. There are some differences such as using camel case or the fact that we pass a function, but overall it is pretty similar.

```
const theLogoIsClicked = () => alert('Clicked');

<Logo onClick={ theLogoIsClicked } />
<input
  type='text'
  onChange={event => theInputIsChanged(event.target.value) } />
```

Using The Bind Function

Usually, we handle events in the component that contains the elements dispatching the events. Like in the example below, we have a click handler and we want to run a function or a method of the same component:

```
class Switcher extends React.Component {
  render() {
    return (
      <button onClick={ this._handleButtonClick }>
        click me
      </button>
    );
  }
  _handleButtonClick() {
    console.log('Button is clicked');
  }
};
```

That's all fine because `_handleButtonClick` is a function and we do pass a function to the `onClick` attribute. The problem, however, is that, as it is, the code doesn't keep context. So, if we have to use `this` inside `_handleButtonClick` to refer to the current `Switcher` component, we will get an error.

```
class Switcher extends React.Component {
  constructor(props) {
    super(props);
    this.state = { name: 'React in patterns' };
  }
  render() {
    return (
      <button onClick={ this._handleButtonClick }>
        click me
      </button>
    );
  }
  _handleButtonClick() {
    console.log(`Button is clicked inside ${ this.state.name }`);
    // leads to
    // Uncaught TypeError: Cannot read property 'state' of null
  }
};
```

What we normally do is to use `bind`:

```
<button onClick={ this._handleButtonClick.bind(this) }>
  click me
</button>
```

However, this means that the `bind` function is called again and again because we may render the button many times. A better approach would be to create the bindings in the constructor of the component:

```
class Switcher extends React.Component {
```

```
  constructor(props) {
    super(props);
    this.state = { name: 'React in patterns' };

    this._buttonClick = this._handleButtonClick.bind(this);
  }
  render() {
    return (
      <button onClick={ this._buttonClick }>
        click me
      </button>
    );
  }
  _handleButtonClick() {
    console.log(`Button is clicked inside ${ this.state.name }`);
  }
};
```

Facebook, by the way, recommends the same technique while dealing with functions that need the context of the same component.

The constructor is also a nice place for partially executing our handlers. For example, we have a form but want to handle every input in a single function.

```
class Form extends React.Component {
  constructor(props) {
    super(props);
    this._onNameChanged = this._onFieldChange.bind(this, 'name');
    this._onPasswordChanged =
      this._onFieldChange.bind(this, 'password');
  }
  render() {
    return (
      <form>
        <input onChange={ this._onNameChanged } />
        <input onChange={ this._onPasswordChanged } />
      </form>
    );
  }
  _onFieldChange(field, event) {
    console.log(`${ field } changed to ${ event.target.value }`);
  }
};
```

Final thoughts

Since we are mostly using HTML-like syntax, it makes sense that we also have DOM-like event handling. Consequently, there is not much additional information to learn about event handling in React since the authors of the library have done an excellent job of keeping what's already there. The following code shows how the event handling code would fit in the complete Form component:

```
import React from 'react';

export default class Form extends React.Component {
  constructor(props) {
    super(props);
    this._onNameChanged = this._onFieldChange.bind(this, 'name');
    this._onPasswordChanged = this._onFieldChange.bind(this, 'password');
  }
  render() {
    return (
      <form>
        <input onChange={ this._onNameChanged } />
        <input onChange={ this._onPasswordChanged } />
      </form>
    );
  }
  _onFieldChange(field, event) {
    console.log(`${ field } changed to ${ event.target.value }`);
  }
};
```

Quick quiz on Event Handlers!

1 Why do we need to bind functions?

COMPLETED 0%

1 of 3

Now that we understand event handling with React components, we will take a closer look at what the components themselves are composed of.

Composition

How components in React can be combined to work together.

WE'LL COVER THE FOLLOWING

- Why Composition Seems like a Good Idea
- Why it's Not The Best Idea
- Using React's children API
- Passing a child as a prop
- Quick quiz on React Composition!

Why Composition Seems like a Good Idea

One of the biggest benefits of React is composability. I personally don't know of any other framework that offers such an easy way to create and combine components. In this section, we will explore a few composition techniques which proved to work well.

Let's look at a simple example. Let's say that we have an application with a header and we want to place navigation inside. We have three React components - `App`, `Header` and `Navigation`. They have to be nested into each other so we end up with the following dependencies:

```
<App> -> <Header> -> <Navigation>
```

The trivial approach for combining these components is to reference them in the places where we need them as we have done in previous lessons.

```
// app.js
import Header from './Header.js';

export default function App() {
  // 'App' referencing 'Header'
  return <Header />;
```

```
}

// Header.js

import Navigation from './Navigation.js';

export default function Header() {
  // 'Header' referencing 'Navigation'
  return <header> <Navigation /> </header>;
}

// Navigation.js
export default function Navigation() {
  return (<nav> ... </nav>);
}
```

Why it's Not The Best Idea

However, by following this approach, we may encounter a couple of problems:

- **Dependencies:** We may consider the `App` as a place where we do our main composition. The `Header`, however, may have other elements like a logo, search field or a slogan. What if we need the same `Header` component, but without the `Navigation`? We can't easily achieve that because we have the two bound tightly together. It would, therefore, be nice if they could instead somehow be passed from the `App` component so we don't create hard-coded dependencies.
- **It's difficult to test**: We may have some business logic in the `Header` and in order to test it, we have to create an instance of the component. However, because it imports other components we will probably create instances of those components too and it becomes difficult to test. Additionally, we could also end up breaking our `Header` test by doing something wrong in the `Navigation` component, which is misleading. *(Note: to some extent shallow rendering solves this problem by rendering only the `Header` without its nested children.)*

Using React's children API

In React we have the handy children prop. That's how the parent reads/accesses its children. This API will make our Header agnostic and dependency-free:

```
import React from 'react';
```

```
export default function Navigation({ children }) {
  return <p> "Some navigation" </p>;
};
```

Notice also that if we don't use `{ children }` in `Header`, the `Navigation` component will never be rendered.

It now becomes easier to test because we may render the `Header` with an empty `<div>`. This will isolate the component and will let us focus on one piece of our application.

Passing a child as a prop

Every React component receives props. As we mentioned already there is no any strict rule about what these props are. We may even pass other components.

```
import React from 'react';

const Title = function () {
  return <h1>Hello there!</h1>;
}
```

This technique is useful when a component like `Header` needs to take decisions about its children, but don't bother about what they actually are. A simple example is a visibility component that hides its children based on a specific condition.

Quick quiz on React Composition!

1 What is composition in React?

In the following two sections, we will describe in detail two methods of composing components in React.

Higher Order Component

Using higher order components as a means to enhance React elements.

WE'LL COVER THE FOLLOWING

- Higher-order components

Higher-order components

For a long period of time, higher-order components were the most popular way to enhance and compose React elements. They look really similar to the decorator design pattern because we have component wrapping and enhancing.

On the technical side, the higher-order component is usually a function that accepts our original component and returns an enhanced/populated version of it. The most trivial example is as follows:

```
var enhanceComponent = (Component) =>
  class Enhance extends React.Component {
    render() {
      return (
        <Component {...this.props} />
      )
    }
  };

var OriginalTitle = () => <h1>Hello world</h1>;
var EnhancedTitle = enhanceComponent(OriginalTitle);

class App extends React.Component {
  render() {
    return <EnhancedTitle />;
  }
};
```

The very first thing that the higher-order component does is to render the original component. It's a good practice to proxy pass the `props` to it. This way we will keep the input of our original component. And here comes the first big

We will keep the input of our original component. And here comes the first big benefit of this pattern - because we control the input of the component we

may send something that the component usually has no access to. Let's say that we have a configuration setting that `OriginalTitle` needs:

```
var config = require('path/to/configuration');

var enhanceComponent = (Component) =>
  class Enhance extends React.Component {
    render() {
      return (
        <Component
          {...this.props}
          title={ config.appTitle }
        />
      )
    }
  };

var OriginalTitle  = ({ title }) => <h1>{ title }</h1>;
var EnhancedTitle = enhanceComponent(OriginalTitle);
```

The knowledge for the `appTitle` is hidden into the higher-order component. `OriginalTitle` knows only that it receives a `prop` called `title`. It has no idea that this is coming from a configuration file. That's a huge advantage because it allows us to isolate blocks. It also helps with the testing of the component because we can create mocks easily.

Another characteristic of this pattern is that we have a nice buffer for additional logic. For example, if our `OriginalTitle` needs data from a remote server as well. We may query this data in the higher-order component and again send it as a prop.

```
var enhanceComponent = (Component) =>
  class Enhance extends React.Component {
    constructor(props) {
      super(props);

      this.state = { remoteTitle: null };
    }
    componentDidMount() {
      fetchRemoteData('path/to/endpoint').then(data => {
        this.setState({ remoteTitle: data.title });
      });
    }
    render() {
      return (
        <Component
          {...this.props}
          title={ config.appTitle }
```

```
          remoteTitle={ this.state.remoteTitle }
        />
      )

    }
  };

var OriginalTitle  = ({ title, remoteTitle }) =>
  <h1>{ title }{ remoteTitle }</h1>;
var EnhancedTitle = enhanceComponent(OriginalTitle);
```

Again, the `OriginalTitle` knows that it receives two props and has to render them next to each other. Its only concern is how the data looks, not where it comes from and how.

Dan Abramov made a really good point that the actual creation of the higher-order component (i.e. calling a function like `enhanceComponent`) should happen at a component definition level. Or in other words, it's a bad practice to do it inside another React component because it may be slow and lead to performance issues.

Given below is an implementation of a higher order component:

```
import React from 'react';
import PropTypes from 'prop-types';
import enhanceComponent from './enhanceComponent.jsx';


var Content = (props) => <p>I am { props.name }</p>;
var EnhancedContent = enhanceComponent(Content);

Content.propTypes = {
  name: PropTypes.string
};

export default class App extends React.Component {
  render() {
    return <EnhancedContent name='Content component' />;
  }
};
```

Now that we know how higher order components work, in the next section, we will be discussing an alternate method of composing components in React.

Functions as Children and the Render Prop Patterns

Patterns that entail using JSX expressions within React to abstract code and provide flexibility.

WE'LL COVER THE FOLLOWING

- JSX Expressions as Children
  - Data Provider
  - Authorize Component
- Quick quiz on these Patterns!
  - Final thoughts

JSX Expressions as Children

Over the last few months, the React community has started shifting in an interesting direction. So far in our examples, the `children` prop was a React component. There is, however, a new pattern gaining popularity in which the same `children` prop is a JSX expression. Let's start by passing a simple object.

```
import React from 'react';

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

This may look weird but in fact is really powerful. Like for example when we have some knowledge in the parent component and don't necessarily want to send it down to children.

The example below prints a list of TODOs. The `App` component has all the data and knows how to determine whether a TODO is completed or not and makes the ones that are bold. The `TodoList` component simply encapsulates the

the ones that are bold. The `TodoList` component simply encapsulates the needed HTML markup.

```
import React from 'react';

function TodoList({ todos, children }) {
  return (
    <section className='main-section'>
      <ul className='todo-list'>{
        todos.map((todo, i) => (
          <li key={ i }>{ children(todo) }</li>
        ))
      }</ul>
    </section>
  );
}

export default TodoList;
```

Notice how the `App` component doesn't expose the structure of the data. `TodoList` has no idea that there is `label` or `status` properties.

The so-called *render prop* pattern is almost the same except that we use the `render` prop and not `children` for rendering the todo.

```
function TodoList({ todos, render }) {
  return (
    <section className='main-section'>
      <ul className='todo-list'>{
        todos.map((todo, i) => (
          <li key={ i }>{ render(todo) }</li>
        ))
      }</ul>
    </section>
  );
}

return (
  <TodoList
    todos={ todos }
    render={
      todo => isCompleted(todo) ?
        <b>{ todo.label }</b> : todo.label
    } />
);
```

These two patterns, *functions as children* and *render prop*, are my favorite ones since recently. They provide flexibility and help when we want to reuse code. They are also a powerful way to abstract imperative code.

```
import React from 'react';
import DataProvider from './dataprovider.js';
```

```
export default class App extends React.Component {
  render() {
    return (
      <DataProvider render={ data => <p>The data is here!</p> } />
    );
  }
}
```

Data Provider

`DataProvider` renders nothing when it first gets mounted. Five seconds later we update the state of the component and render a `<section>` followed by what the `render` prop is returning. Imagine that this same component fetches data from a remote server and we want to display it only when it is available.

```
<DataProvider render={ data => <p>The data is here!</p> } />
```

Authorize Component

We do say what we want to happen but not how. That is hidden inside the `DataProvider`. These days we use this pattern at work where we have to restrict some UI to certain users who have `read:products` permissions. And we used the *render prop* pattern.

```
<Authorize
  permissionsInclude={[ 'read:products' ]}
  render={ () => <ProductsList /> } />
```

This should be self-explanatory. `Authorize` goes to our identity provider and checks what are the permissions of the current user. If he/she is allowed to read our products we render the `ProductList`.

Quick quiz on these Patterns!

1 What is an advantage of the *functions as children* pattern?

Final thoughts

Have you wondered why HTML is still popular? It was created in the dawn of the internet and we still use it. That is because it's highly composable. React and its JSX looks like HTML on steroids and as such, it comes with the same capabilities. So, make sure that you master composition because it is one of the greatest benefits of React.

Controlled and Uncontrolled Inputs

How inputs can be handled in React.

WE'LL COVER THE FOLLOWING

- Controlled Inputs
- Uncontrolled Input
- Final thoughts
- Quick Quiz on Controlled and Uncontrolled Inputs!

The two terms *controlled* and *uncontrolled* are very often used in the context of forms management.

Controlled Inputs

A *controlled* input is an input that gets its value from a single source of truth. For example, the `App` component below has a single `<input>` field which is *controlled*:

```
class App extends React.Component {
  constructor(props) {
    super(props);
    this.state = { value: 'hello' };
  }
  render() {
    return <input type='text' value={ this.state.value } />;
  }
};
```

The result of this code is an input element that we can focus, but can't change. It is never updated because we have a single source of truth - the `App`'s component state.

To make the input work as expected we have to add an `onChange` handler and update the state (the single source of truth), which will trigger a new

rendering cycle and we will see what we typed.

```
class App extends React.Component {
  constructor(props) {
    super(props);
    this.state = { value: 'hello' };
    this._change = this._handleInputChange.bind(this);
  }
  render() {
    return (
      <input
        type='text'
        value={ this.state.value }
        onChange={ this._change } />
    );
  }
  _handleInputChange(e) {
    this.setState({ value: e.target.value });
  }
};
```

Uncontrolled Input

On the other hand, we have *uncontrolled* input, where we let the browser handle the user's updates. We may still provide an initial value by using the `defaultValue` prop but after that, the browser is responsible for keeping the state of the input.

```
class App extends React.Component {
  constructor(props) {
    super(props);
    this.state = { value: 'hello' };
  }
  render() {
    return <input type='text' defaultValue={ this.state.value } />
  }
};
```

That `<input>` element above is a little bit useless because the user updates the value but our component has no idea about that. We then have to use `Refs` to get access to the actual element.

```
class App extends React.Component {
  constructor(props) {
    super(props);
    this.state = { value: 'hello' };
    this._change = this._handleInputChange.bind(this);
  }
  render() {
    return (
      <input
```

```
      <input
        type='text'
        defaultValue={ this.state.value }
        onChange={ this._change }
        ref={ input => this.input = input }/>
    );
  }
  _handleInputChange() {
    this.setState({ value: this.input.value });
  }
};
```

The `ref` prop receives a string or a callback. The code above uses a callback and stores the DOM element into a *local* variable called `input`. Later when the `onChange` handler is fired we get the new value and send it to the `App`'s state.

Using a lot of `refs` is not a good idea. If it happens in your app consider using `controlled` inputs and re-think your components.

Final thoughts

The choice between *controlled* and *uncontrolled* inputs is very often underrated. However, I believe that it is a fundamental decision because it dictates the data flow in the React component. Given below is code for how uncontrolled inputs would be handled. However, I personally think that *uncontrolled* inputs are kind of an anti-pattern and would avoid them when possible.

```
import React from 'react';
require('./style.css');

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

Now that we have discussed the different types of inputs React components can take, we will proceed to look into the different types of components one can have created in React in the following section.

Quick Quiz on Controlled and Uncontrolled Inputs!

Q

When we update our value from a single source of truth, it is an example of uncontrolled input.

Presentational and Container Components

Splitting React components into presentations and containers to manage state.

WE'LL COVER THE FOLLOWING

- The problems
- Extracting the container
- Presentational component
- Benefits
- Final thoughts
- Quick Quiz on Presentational and Container Components!

Every beginning is difficult. React is no exception and as beginners, we also have lots of questions. Where am I supposed to put my data, how to communicate changes or how to manage state? The answers to these questions are very often a matter of context and sometimes just practice and experience with the library. However, there is a pattern which is used widely and helps to organize React based applications - splitting the component into presentation and container.

Let's start with a simple example that illustrates the problem and then split the component into a container and presentation. We will use a `Clock` component. It accepts a Date object as a prop and displays the time in real time.

```
class Clock extends React.Component {
  constructor(props) {
    super(props);
    this.state = { time: this.props.time };
    this._update = this._updateTime.bind(this);
  }
  render() {
    const time = this._formatTime(this.state.time);
    return (
      <h1>
```

```
        { time.hours } : { time.minutes } : { time.seconds }
      </h1>
    );
  }
  componentDidMount() {
    this._interval = setInterval(this._update, 1000);
  }
  componentWillUnmount() {
    clearInterval(this._interval);
  }
  _formatTime(time) {
    var [ hours, minutes, seconds ] = [
      time.getHours(),
      time.getMinutes(),
      time.getSeconds()
    ].map(num => num < 10 ? '0' + num : num);

    return { hours, minutes, seconds };
  }
  _updateTime() {
    this.setState({
      time: new Date(this.state.time.getTime() + 1000)
    });
  }
};

ReactDOM.render(<Clock time={ new Date() }/>, ...);
```

In the constructor of the component, we initialize the component's state which in our case is just the current `time` value. By using `setInterval` we update the state every second and the component is re-rendered. To make it look like a real clock we use two helper methods - `_formatTime` and `_updateTime`. The former extracts hours, minutes and seconds and makes sure that they are following the two-digit format. `_updateTime` mutates the current `time` object by one second.

The problems

There are a couple of things happening in our component. It looks like it has too many responsibilities.

- It mutates the state by itself. Changing the time inside the component may not be a good idea because then only `Clock` knows the current value. If there is another part of the system that depends on this data it will be difficult to share it.
- `_formatTime` is actually doing two things - it extracts the needed information from the date object and makes sure that the values are always presented by two digits. That's fine but it will be nice if the extracting is not part of the function because then it is bound to the type

extracting is not part of the function because then it is bound to the type

of the time object (coming as a prop), i.e., knows specifics about the shape of the data and at the same time deals with the visualization of it.

Extracting the container

Containers know about data, its shape and where it comes from. They know details about how the things work or the so-called *business logic*. They receive information and format it so that it is easy to be used by the presentational component. Very often we use higher-order components to create containers because they provide a buffer space where we can insert custom logic.

Here's what our `ClockContainer` looks like:

```
// Clock/index.js
import Clock from './Clock.jsx'; // <-- that's the presentational component

export default class ClockContainer extends React.Component {
  constructor(props) {
    super(props);
    this.state = { time: props.time };
    this._update = this._updateTime.bind(this);
  }
  render() {
    return <Clock { ...this._extract(this.state.time) }/>;
  }
  componentDidMount() {
    this._interval = setInterval(this._update, 1000);
  }
  componentWillUnmount() {
    clearInterval(this._interval);
  }
  _extract(time) {
    return {
      hours: time.getHours(),
      minutes: time.getMinutes(),
      seconds: time.getSeconds()
    };
  }
  _updateTime() {
    this.setState({
      time: new Date(this.state.time.getTime() + 1000)
    });
  }
};
```

It still accepts `time` (a date object), does the `setInterval` loop and knows details about the data (`getHours`, `getMinutes` and `getSeconds`). In the end, it renders the presentational component and passes three numbers for hours,

minutes and seconds. There is nothing about how the things look like. Only *business logic*.

Presentational component

Presentational components are concerned with how the things look. They have the additional markup needed for making the page pretty. Such components are not bound to anything and have no dependencies. Very often implemented as stateless functional components they don't have internal state.

In our case the presentational component contains only the two-digits check and returns the `<h1>` tag:

```
// Clock/Clock.jsx
export default function Clock(props) {
  var [ hours, minutes, seconds ] = [
    props.hours,
    props.minutes,
    props.seconds
  ].map(num => num < 10 ? '0' + num : num);

  return <h1>{ hours } : { minutes } : { seconds }</h1>;
};
```

Benefits

Splitting the components in containers and presentation increases the reusability of the components. Our `Clock` function/component may exist in an application that doesn't change the time or it's not working with JavaScript's Date object. That's because it is pretty *dummy*. No details about the data are required.

The containers encapsulate logic and we may use them together with different renderers because they don't leak information about the visual part. The approach that we took above is a good example of how the container doesn't care about how the things look like. We may easily switch from digital to an analog clock and the only one change will be to replace the `<Clock>` component in the `render` method.

Even the testing becomes easier because the components have fewer responsibilities. Containers are not concerned with UI. Presentational components are pure renderers and it is enough to run expectations on the resulted markup.

resulted markup.

Final thoughts

The concept of container and presentation is not new at all but it fits really nicely with React. It makes our applications better structured, easy to manage and scale. Given below is the complete working example of how containers and presentations achieve this:

```
import React from 'react';
import PropTypes from 'prop-types';

export default function Clock(props) {
  var [ hours, minutes, seconds ] = [
    props.hours,
    props.minutes,
    props.seconds
  ].map(num => num < 10 ? '0' + num : num);

  return <h1>{ hours } : { minutes } : { seconds }</h1>;
};
Clock.propTypes =
  {
  hours: PropTypes.number.isRequired,
  minutes: PropTypes.number.isRequired,
  seconds: PropTypes.number.isRequired
};
```

Quick Quiz on Presentational and Container Components!

Q

```
export default function Calendar(props) {
  var [ date, month, year ] = [
    props.date,
    props.month,
    props.year
  ]

  return <h1>{ day } : { month } : { year }</h1>;
};
```

The code above is an example of which of the following?

One-Way Direction Data Flow

This lesson will explain with an example how and why React components avoid modifying the data that they receive and instead look for changes in this data.

WE'LL COVER THE FOLLOWING

- The Idea
- Problem with Data Flow
- Solution: One-way Data Flow
- Final thoughts
- Quick Quiz

The Idea

One-way direction data flow is the idea that the components do not modify the data that they receive. They only listen for changes in this data and maybe provide the new value but they do not update the actual data. This update happens following another mechanism, in another place and the component just gets re-rendered with the new value, if any.

Problem with Data Flow

Let's, for example, get a simple `Switcher` component that contains a button. If we click it, we have to enable a flag in the system.

```
import React from 'react';
require('./style.css');

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

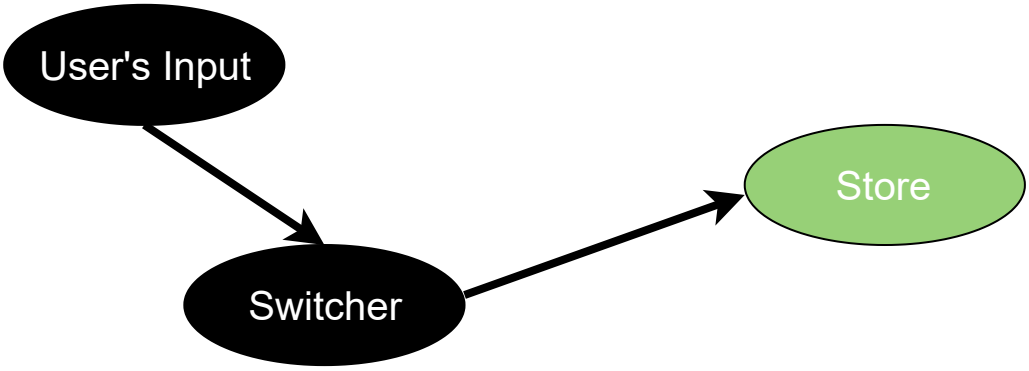
At this moment we have the data inside our component, in the form of `this.state.flag`. Or in other words, `Switcher` is the only one place that knows about our `flag`. Let's send it out to some kind of a store:

```
import React from 'react';
require('./style.css');

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

Our `Store` object is a singleton where we have helpers for setting and getting the value of the `_flag` property. By passing the setter to the `Switcher` we are able to update the data externally. More or less our application workflow looks like that:

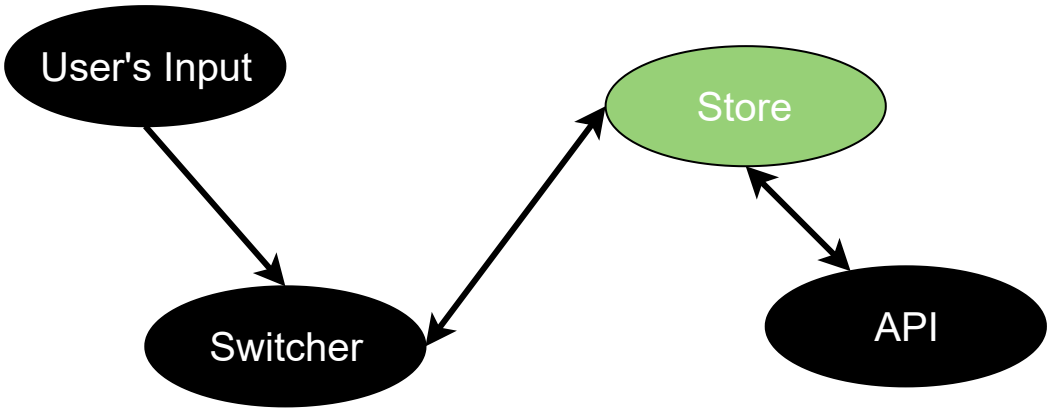


Let's assume that we are saving the flag value to a back-end service via the `Store`. When the user comes back we have to set a proper initial state. If the user left the flag as `true` we have to show *"lights on"* and not the default *"lights off"*. Now it gets tricky because we have the data in two places. The UI and the `Store` have their own states. We have to communicate in both directions from the store to the switcher and from the switcher to the store.

```
// ... in App component
<Switcher
  value={ Store.get() }
  onChange={ Store.set.bind(Store) } />

// ... in Switcher component
constructor(props) {
  super(props);
  this.state = { flag: this.props.value };
  ...
```

Our workflow changes to what is shown the figure below



All this leads to managing two states instead of one. What if the `Store` changes its value based on other actions in the system. We have to propagate that change to the `Switcher` and we increase the complexity of our app.

Solution: One-way Data Flow

One-way direction data flow solves this problem. It eliminates the multiple places where we manage states and deals with the only one which is usually the store. To achieve that we have to tweak our `Store` object a little bit. We need logic that allows us to subscribe to changes:

```
// this store has getter, setter routines for
// state variables
var Store = {
  _handlers: [],
  _flag: '',
  subscribe: function(handler) {
    this._handlers.push(handler);
  },
  set: function(value) {
    this._flag = value;
    this._handlers.forEach(handler => handler(value))
  },
  get: function() {
    return this._flag;
  }
};
```

Then we will hook our main `App` component and we'll re-render it every time when the `Store` changes its value:

```
class App extends React.Component {
  constructor(props) {
    super(props);
    // get current state value
    this.state = { value: Store.get() };
```

```
    Store.subscribe(value => this.setState({ value }));
  }
  render() {

    return (
      <div>
        <Switcher
          value={ this.state.value }
          onChange={ Store.set.bind(Store) } />
      </div>
    );
  }
};
```

Because of this change, the `Switcher` becomes really simple. We don't need the internal state and the component may be written as a stateless function.

```
function Switcher({ value, onChange }) {
  return (
    <button onClick={ e => onChange(!value) }>
      { value ? 'lights on' : 'lights off' }
    </button>
  );
};

<Switcher
  value={ Store.get() }
  onChange={ Store.set.bind(Store) } />
```

Final thoughts

The following example shows how this pattern can be incorporated in to a React application:

```
import React from 'react';
require('./style.css');

import ReactDOM from 'react-dom';
import App from './app.js';

ReactDOM.render(
  <App />,
  document.getElementById('root')
);
```

The benefit that comes with this pattern is that our components become a dummy representation of the store's data. There is only one source of truth and this makes development easier. If you are going to take one thing from this course, it should be the concept of one-directional data flow. It's that important in React.

Quick Quiz

1 `App` component re-renders every time the `store` changes its value:

COMPLETED 0%

1 of 2

In the next lesson, we will start off with Flux, an architectural design pattern for building user interfaces, introduced by Facebook!

Architecture and Main Characteristics

Introducing Flux, its architectural design pattern for building user interfaces, commonly used with React to make code simpler and its main characteristics.

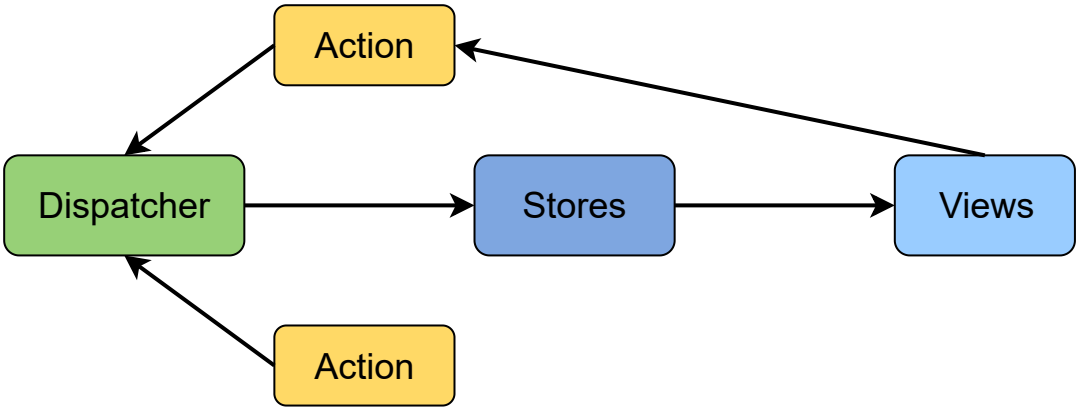
WE'LL COVER THE FOLLOWING

- Flux: What is it?
- Flux: Architecture and Main Characteristics

Flux: What is it?

Flux is an architectural design pattern for building user interfaces. It was introduced by Facebook at their F8 conference. Since then, lots of companies adopted the idea and it seems like a good pattern for building front-end apps. Flux is very often used with React and is simple and really flexible. The pattern helps to create apps faster and at the same time keeps the code well organized.

Flux: Architecture and Main Characteristics



The main actor in this pattern is the *dispatcher*. It acts as a hub for all the events in the system. Its job is to receive notifications that we call *actions* and pass them to all the *stores*. The store decides if it is interesting or not and reacts by changing its internal state/data. That change is triggering re-rendering of the *views* which are (in our case) React components. If we have

rendering of the *views* which are (in our case) React components. If we have

to compare Flux to the well known MVC we may say that the store is similar to the model. It keeps the data and its mutations.

The actions are coming to the dispatcher either from the views or from other parts of the system, like services. For example, a module that performs a HTTP request. When it receives the result it may fire an action saying that the request was successful.

Now that we know what Flux is, we can move on to see how a Flux architecture can be implemented. As every other popular concept Flux also has some variations. Very often to understand something we have to implement it. In the next few lessons, we will create a library that provides helpers for building the Flux pattern.

The Dispatcher

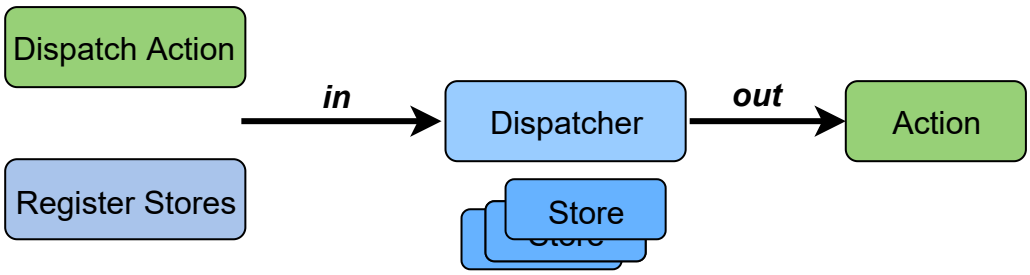
In this lesson, we will look at the Dispatcher as the main component in the Flux architecture, and some code related to it in detail.

WE'LL COVER THE FOLLOWING

- The Dispatcher

The Dispatcher

In most cases, we need a single dispatcher because it acts as a sort of glue for the rest of the parts and it, therefore, makes sense to have only one. The dispatcher needs to know two things - actions and stores. The actions are simply forwarded to the stores so we don't necessarily have to keep them. The stores, however, should be tracked inside the dispatcher so we can loop through them:



Following is what we can start with:

```
var Dispatcher = function () {
  return {
    _stores: [],
    register: function (store) {
      this._stores.push({ store: store });
    },
    // checks for updates in each store
    dispatch: function (action) {
      if (this._stores.length > 0) {
        this._stores.forEach(function (entry) {
          entry.store.update(action);
        });
```

```
      }
    }
  }
};
```

The first thing that we notice is that we *expect* to see an `update` method in the passed stores. It will be nice to throw an error if such method is not there:

```
// throwing an error if no expected update method exists
register: function (store) {
  if (!store || !store.update) {
    throw new Error('You should provide a store that has an `update` method.');
  } else {
    this._stores.push({ store: store });
  }
}
```

Let’s now move on to Views, Stores and their bounding

Bounding Views and Stores

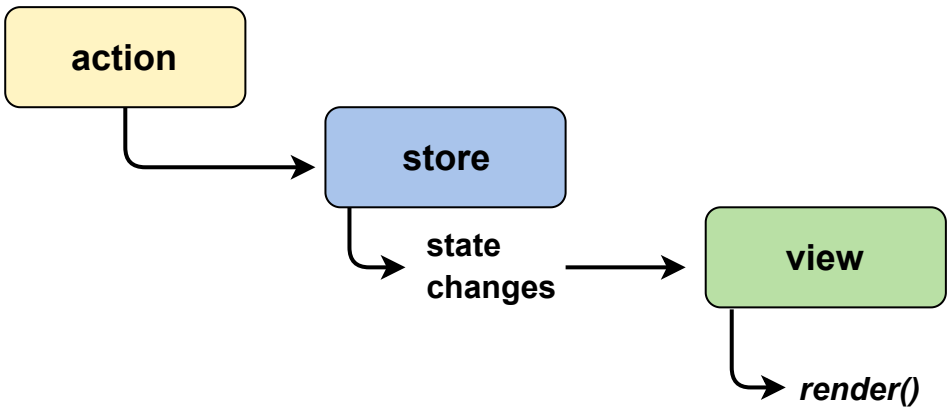
This lesson introduces multiple ways to connect views to stores

WE'LL COVER THE FOLLOWING

- Bounding the views and the stores
  - Using a helper
  - With a mixin
  - Using a context
  - Higher-Order components concept
  - Using Higher Order Components

Bounding the views and the stores

The next logical step is to connect our views to the stores so we re-render when the state in the stores is changed.



There are multiple ways to achieve this. Let's look at a few:

Using a helper

Some of the flux implementations available out there provide a helper function that does the job. For example:

```
Framework.attachToStore(view, store);
```

However, to make `attachToStore` work we expect to see a specific API in the view and in the store. We kind of strictly define new public methods. Or in other words, we say “Your views and store should have such APIs so we are able to wire them together”. If we go down this road then we will probably define our own base classes which could be extended so we don’t bother the developer with Flux details. Then we say “All your classes should extend our classes”. This doesn’t sound good either because the developer may decide to switch to another Flux provider and has to amend everything.

With a mixin

What if we use React’s mixins? Like this:

```
var View = React.createClass({
  mixins: [Framework.attachToStore(store)]
  ...
});
```

That’s a nice way to define the behavior of an existing React component. So, in theory, we may create a mixin that does the bounding for us. Mixins modify the components in a non-predictable way. The developer has no idea what is going on behind the scenes.

Using a context

Another technique that may answer the question is React’s context. It is a way to pass props to child components without the need to specify them in every level of the tree. Facebook suggests a context in the cases where we have data that has to reach deeply nested components.

> Occasionally, you want to pass data through the component tree without having to pass the props down manually at every level. React’s “context” feature lets you do this.

You might see a similarity with the mixins here. The context is defined somewhere at the top and magically serves props for all the children below. It's not immediately clear where the data comes from.

Higher-Order components concept

Higher-Order components pattern is introduced by Sebastian Markbåge and it is about wrapping a component and attaching some new functionalities or props to it. For example:

```
// creating a wrapper component that may send props
// and apply additional logic
function attachToStore(Component, store, consumer) {
  const Wrapper = React.createClass({
    getInitialState() {
      return consumer(this.props, store);
    },
    componentDidMount() {
      store.onChangeEvent(this._handleStoreChange);
    },
    componentWillUnmount() {
      store.offChangeEvent(this._handleStoreChange);
    },
    _handleStoreChange() {
      if (this.isMounted()) {
        this.setState(consumer(this.props, store));
      }
    },
    render() {
      return <Component {...this.props} {...this.state} />;
    }
  });
  return Wrapper;
};
```

`Component` is the view that we want to be attached to the `store`. The `consumer` function says what part of the store's state should be fetched and sent to the view. A simple usage of the above function could be:

```
class MyView extends React.Component {
  ...
}

ProfilePage = connectToStores(MyView, store, (props, store) => ({
  data: store.get('key')
}));
```

That is an interesting pattern because it shifts the responsibilities. It is the view fetching data from the store and not the store pushing something to the view. This of course has it's own pros and cons. It is nice because it makes the store dummy. A store that only mutates the data and says "Hey, my state is changed". It is not responsible for sending anything to anyone.

The downside of this approach is maybe the fact that we have one more component (the wrapper) involved. We also need the three things - view, store, and consumer to be in one place so we can establish the connection.

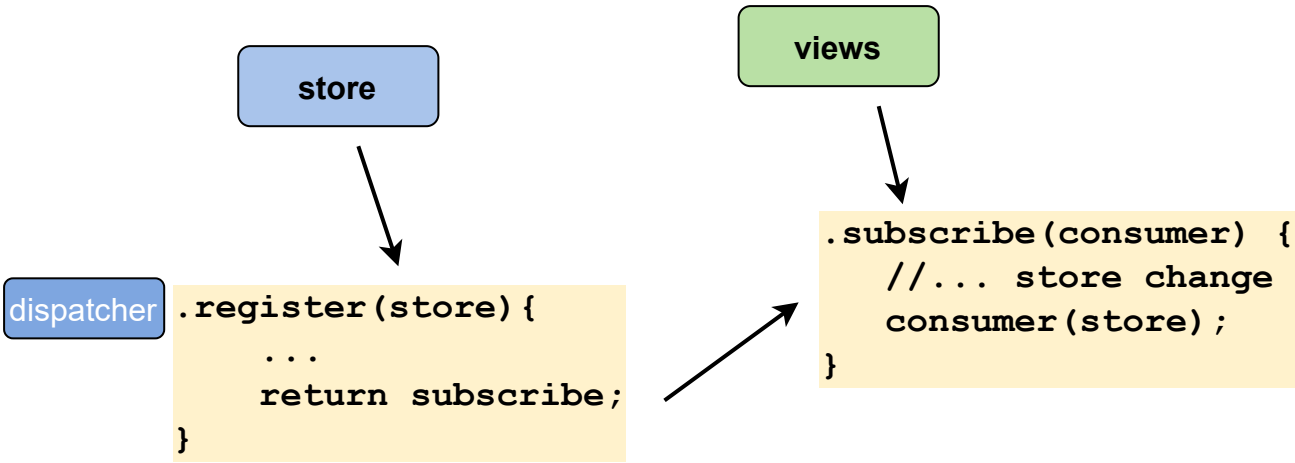
Using Higher Order Components

In higher-order components, the view decides what it needs. That *knowledge* anyway exists in the component so it makes sense to keep it there. That's also why the functions that generate higher-order components are usually kept in the same file as the view. What if we can use a similar approach but not passing the store at all. Or in other words, a function that accepts only the consumer. And that function is called every time when there is a change in the store.

So far our implementation interacts with the store only in the `register` method.

```
register: function (store) {
  if (!store || !store.update) {
    throw new Error('You should provide a store that has an `update` metho
d.');
  } else {
    this._stores.push({ store: store });
  }
}
```

By using `register` we keep a reference to the store inside the dispatcher. However, `register` returns nothing. And instead of nothing, it may return a **subscriber** that will accept our consumer functions.



Connect store and view

Let's say we decided to send the whole store to the consumer function and not the data that the store keeps. Like in the higher-order components pattern the view should say what it needs by using the store's getters. This makes the store really simple and there is no trace of presentational logic.

Here is what the register method looks like after the changes:

```
register: function (store) {
  // throw error if no update method exists
  if (!store || !store.update) {
    throw new Error(
      'You should provide a store that has an `update` method.'
    );
  } else {
    var consumers = [];
    var subscribe = function (consumer) {
      consumers.push(consumer);
    };

    this._stores.push({ store: store });
    return subscribe;
  }
  return false;
}
```

The last bit in the story is how the store says that its internal state is changed. It's nice that we collect the consumer functions but right now there is no code that executes them.

According to the basic principles of the flux architecture the stores change their state in response of actions. In the `update` method we send the `action` but we could also send a function `change`. Calling that function should trigger

the consumers:

```
register: function (store) {
  // throw error if no update method exists
  if (!store || !store.update) {
    throw new Error(
      'You should provide a store that has an `update` method.'
    );
  } else {
    var consumers = [];
    var change = function () {
      consumers.forEach(function (consumer) {
        consumer(store);
      });
    };
    var subscribe = function (consumer) {
      consumers.push(consumer);
    };

    this._stores.push({ store: store, change: change });
    return subscribe;
  }
  return false;
},
dispatch: function (action) {
  if (this._stores.length > 0) {
    this._stores.forEach(function (entry) {
      entry.store.update(action, entry.change);
    });
  }
}
```

Notice how we push `change` together with `store` inside the `_stores` array. Later in the `dispatch` method, we call `update` by passing the `action` and the `change` function.

A common use case is to render the view with the initial state of the store. In the context of our implementation, this means firing all the consumers at least once when they land in the library. This could be easily done in the `subscribe` method:

```
var subscribe = function (consumer, noInit) {
  consumers.push(consumer);
  !noInit ? consumer(store) : null;
};
```

Of course, sometimes this is not needed so we added a flag which is by default false. Here is the final version of our dispatcher:

```
var Dispatcher = function () {
  return {

    _stores: [],
    register: function (store) {
      // throw error if no update method exists
      if (!store || !store.update) {
        throw new Error(
          'You should provide a store that has an `update` method'
        );
      } else {
        var consumers = [];
        var change = function () {
          consumers.forEach(function (consumer) {
            consumer(store);
          });
        };
        var subscribe = function (consumer, noInit) {
          consumers.push(consumer);
          !noInit ? consumer(store) : null;
        };

        this._stores.push({ store: store, change: change });
        return subscribe;
      }
      return false;
    },
    // check all stores for update
    dispatch: function (action) {
      if (this._stores.length > 0) {
        this._stores.forEach(function (entry) {
          entry.store.update(action, entry.change);
        });
      }
    }
  }
};
```

We will continue to build upon this in the following section to produce a complete working Flex architecture.

The Actions

This lesson introduces actions and their creation in the Flux architecture

WE'LL COVER THE FOLLOWING

- The actions

The actions

You probably noticed that we didn't talk about the actions. What are they? The convention is that they should be simple objects having two properties - `type` and `payload`:

```
{
  type: 'USER_LOGIN_REQUEST',
  payload: {
    username: '...',
    password: '...'
  }
}
```

The `type` says what exactly the action is and the `payload` contains the information associated with the event. And in some cases, we may leave the `payload` empty.

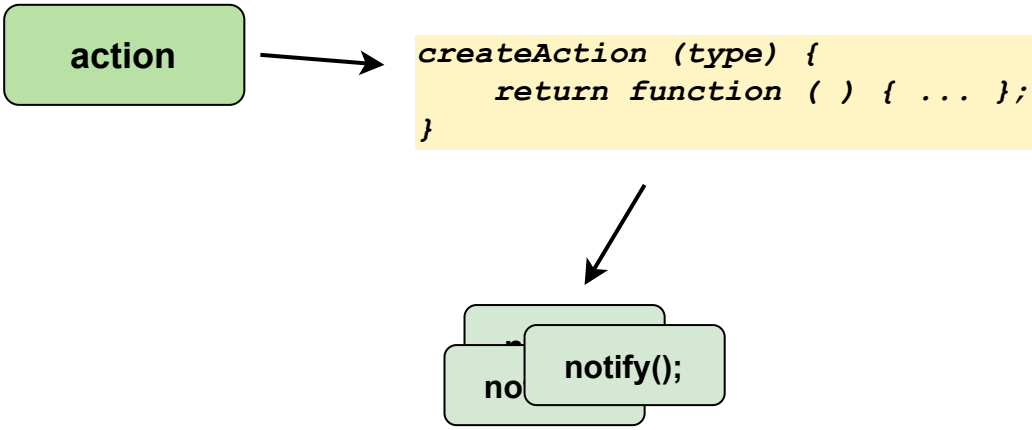
It's interesting that the `type` is well known in the beginning. We know what type of actions should be floating in our app, who is dispatching them and which of the stores are interested. Thus, we can apply partial application and avoid passing the action object here and there. For example:

```
var createAction = function (type) {
  // actions must have a type
  if (!type) {
    throw new Error('Please, provide action\'s type.');
  } else {
    return function (payload) {
```

```
      // pass action to dispatcher
      return dispatcher.dispatch({
        type: type,
        payload: payload
      });
    }
  }
}
```

`createAction` leads to the following benefits:

- We need not to remember the exact type of the action. We now have a function which we call passing only the payload.
- We don't need an access to the dispatcher anymore which is a huge benefit. Otherwise, think about how we have to pass it to every single place where we need to dispatch an action.
- In the end, we don't have to deal with objects but with functions which is much nicer. The objects are *static* while the functions describe a *process*.



Flux Action Creators

This approach for creating actions is actually really popular and functions like the one above are usually called *action creators*.

In the next lesson, we will wrap up the code, presenting the unified version.

Final Code

This lesson unifies the code presented in the last few lessons to give the Flux architecture the final shape

WE'LL COVER THE FOLLOWING

- Full Implementation of Flux Architecture

Full Implementation of Flux Architecture

In the last few lessons, we successfully hid the dispatcher while submitting actions. We may do it again for the store's registration:

```
var createSubscriber = function (store) {
  return dispatcher.register(store);
}
```

And instead of exporting the dispatcher we may export only these two functions `createAction` and `createSubscriber`. Here is how the final code looks like:

```
var Dispatcher = function () {
  return {
    _stores: [],
    register: function (store) {
      // expecting an `update` method for each store
      if (!store || !store.update) {
        throw new Error(
          'You should provide a store that has an `update` method'
        );
      } else {
        var consumers = [];
        var change = function () {
          consumers.forEach(function (consumer) {
            consumer(store);
          });
        };
        var subscribe = function (consumer, noInit) {
          consumers.push(consumer);
          !noInit ? consumer(store) : null;
        };
```

```
        this._stores.push({ store: store, change: change });
        return subscribe;

      }
      return false;
    },
    dispatch: function (action) {
      // check all stores for update
      if (this._stores.length > 0) {
        this._stores.forEach(function (entry) {
          entry.store.update(action, entry.change);
        });
      }
    }
  }
};

module.exports = {
  create: function () {
    var dispatcher = Dispatcher();

    return {
      createAction: function (type) {
        if (!type) {
          throw new Error('Please, provide action\'s type.');
        } else {
          return function (payload) {
            // actions are passed to dispatcher
            return dispatcher.dispatch({
              type: type,
              payload: payload
            });
          }
        }
      },
      createSubscriber: function (store) {
        return dispatcher.register(store);
      }
    }
  }
};
```

If we add the support of AMD, CommonJS and global usage we end up with 66 lines of code, 1.7KB plain or 795 bytes after minification JavaScript.

Let’s wrap up our discussion on Flux in the next lesson

Wrapping Up

This lesson brings all the little components we have learned in the previous lessons together, and presents the final product

WE'LL COVER THE FOLLOWING

- The markup
- The view
- The store
- Wiring all the pieces
- A live demo
- Quick Quiz on Flux:

Let's write a simple counter app that doesn't involve React so we see the pattern in action.

The markup

We'll need some UI to interact with it so:

```
<!-- Simple HTML document with `increase` and `decrease` buttons -->
html
<div id="counter">
  <span></span>
  <button>increase</button>
  <button>decrease</button>
</div>
```

The `span` will be used for displaying the current value of our counter. The buttons will change that value.

The view

```
const View = function (subscribeToStore, increase, decrease) {
  var value = null;
```

```
  var el = document.querySelector('#counter'); // selecting div by ID
  var display = el.querySelector('span'); // value displayed here
  var [ increaseBtn, decreaseBtn ] =

    Array.from(el.querySelectorAll('button'));

  var render = () => display.innerHTML = value; // render value
  var updateState = (store) => value = store.getValue(); // update state

  // subscribe state to store
  subscribeToStore([updateState, render]);

  // listen to button clicks
  increaseBtn.addEventListener('click', increase);
  decreaseBtn.addEventListener('click', decrease);
};
```

It accepts a store subscriber function and two action function for increasing and decreasing the value. The first few lines of the view are just fetching the DOM elements.

After that, we define a `render` function which puts the value inside the `span` tag. `updateState` will be called every time when the store changes. So, we pass these two functions to `subscribeToStore` because we want to get the view updated and we want to get an initial rendering. Remember how our consumers are called at least once by default.

The last bit is calling the action functions when we press the buttons.

The store

Every action has a type. It's a good practice to create constants for these types so we don't deal with raw strings.

```
const INCREASE = 'INCREASE';
const DECREASE = 'DECREASE';
```

Very often we have only one instance of every store. For the sake of simplicity, we'll create ours as a singleton.

```
// returns updated value on state change
const CounterStore = {
  _data: { value: 0 },
  getValue: function () {
    return this._data.value;
  },
  update: function (action, change) {
    if (action.type === INCREASE) {
      this._data.value += 1;
    } else if (action.type === DECREASE) {
```

```
    } else if (action.type === DECREASE) {
      this._data.value -= 1;
    }
    change();
  }
};
```

`_data` is the internal state of the store. `update` is the well known method that our dispatcher calls. We process the action inside and say `change()` when we are done. `getValue` is a public method used by the view so it reaches the needed info. In our case this is just the value of the counter.

Wiring all the pieces

So, we have the store waiting for actions from the dispatcher. We have the view defined. Let's create the store subscriber, the actions and run everything.

```
const { createAction, createSubscriber } = Fluxiny.create();
const counterStoreSubscriber = createSubscriber(CounterStore);
// create `increase`, `decrease` actions to cause a state change
const actions = {
  increase: createAction(INCREASE),
  decrease: createAction(DECREASE)
};

View(counterStoreSubscriber, actions.increase, actions.decrease);
```

And that's it. Our view is subscribed to the store and it renders by default because one of our consumers is actually the `render` method.

A live demo

A live demo could be seen in the following JSBin. If that's not enough and it seems too simple for you please checkout the example in Fluxiny repository . It uses React as a view layer.

Quick Quiz on Flux:

1 What is the hub for all the events in the Flux architecture?

And that is about it for Flux Architecture. Let’s now move onto Redux.

The Idea

In this lesson we re-visit what is One-way Direction data flow and why we need it

WE'LL COVER THE FOLLOWING

- One-Way Direction Data Flow

One-Way Direction Data Flow

As discussed in the beginning of previous chapter, one-way direction data flow is the idea that the components do not modify the data that they receive. They only listen for changes in this data and might provide the new value but they do not update the actual data. This update happens following another mechanism, in another place and the component just gets re-rendered with the new value, if any.

Feel free to revisit the problem with data flow with example if you don't recall it. In the next lesson, we will start discussing Redux.

Architecture and Main Characteristics

This lesson introduces Redux and its main characteristics

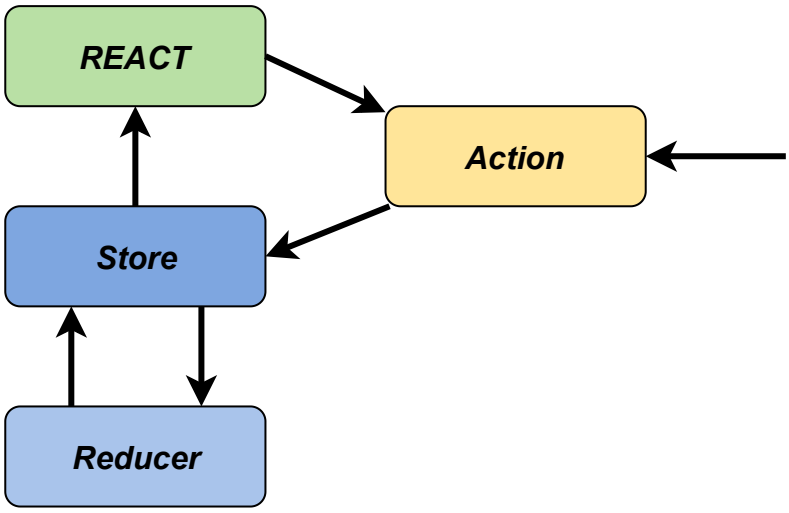
WE'LL COVER THE FOLLOWING

- What is it?
- Architecture and Main Characteristics

What is it?

Redux is a library that acts as a state container and helps to manage your application data flow. It was introduced back in 2015 at ReactEurope conference (video) by Dan Abramov. It is similar to Flux architecture and has a lot in common with it. In this section we will create a small counter app using Redux alongside React.

Architecture and Main Characteristics



Redux Architecture

Similar to Flux architecture we have the view components (React) dispatching an action. Same action may be dispatched by another part of our system. Like a bootstrap logic for example. This action is dispatched not to a central hub but directly to the store. We are saying "store" not "stores" because there is

but directly to the store. We are saying "store" not "stores" because there is only one in Redux. That is one of the big differences between Flux and Redux.

The logic that decided how our data changes lives in pure functions called reducers. Once the store receives an action it asks the reducers about the new version of the state by sending the current state and the given action. Then in immutable fashion the reducer needs to return the new state. The store continues from there and updates its internal state. As a final step, the React component wired to the store gets re-rendered

The concept is pretty linear and again follows the one-direction data flow.

Let's talk about all these pieces and introduce a couple of new terms that support the work of the Redux pattern.

Actions, Store and Reducer

Creating an app using Redux alongside React to demonstrate the similarities and differences between Flux and Redux.

WE'LL COVER THE FOLLOWING

- Actions
- Store
- Reducer

Actions

Like Flux, the typical action in Redux is merely an object with a `type` property. Everything else in that object is considered to be context specific data and it is not related to the pattern, but to your application logic. For example:

```
const CHANGE_VISIBILITY = 'CHANGE_VISIBILITY';
const action = {
  type: CHANGE_VISIBILITY,
  visible: false
}
```

It is a good practice that we create constants like `CHANGE_VISIBILITY` for our action types. It happens that there are lots of tools/libraries that support Redux and solve different problems which do require the type of the action only. So it is just a convenient way to transfer this information.

The `visible` property is the meta data that we mentioned above. It has nothing to do with Redux. It means something in the context of the application.

Every time when we want to dispatch a method we have to use such objects. However, it becomes too noisy to write them over and over again. That is why

there is the concept of *action creators*. An action creator is a function that returns an object and may or may not accept an argument which directly relates to the action properties. For example the action creator for the above action looks like this:

```
const changeVisibility = visible => ({
  type: CHANGE_VISIBILITY,
  visible
});

changeVisibility(false);
// { type: CHANGE_VISIBILITY, visible: false }
```

Notice that we pass the value of the `visible` as an argument and we don't have to remember (or import) the exact type of the action. Using such helpers makes the code compact and easy to read.

Store

Redux provides a helper `createStore` for creating a store. Its signature is as follows:

```
import { createStore } from 'redux';

createStore([reducer], [initial state], [enhancer]);
```

We already mentioned that the reducer is a function that accepts the current state, action and returns the new state. More about that in a bit. The second argument is the initial state of the store. This comes as a handy instrument to initialize our application with data that we already have. This feature is the essence of processes like server-side rendering or persistent experience. The third parameter, enhancer, provides an API for extending Redux with third party middlewares and basically plugs some functionally which is not baked-in. Like an instrument for handling async processes.

Once created, the store has four methods - `getState`, `dispatch`, `subscribe` and `replaceReducer`. Probably the most important one is `dispatch`:

```
store.dispatch(changeVisibility(false));
```

That is the place where we use our action creators. We pass the result to them or, in other words, action objects to this `dispatch` method. It then gets spread

or, in other words, action objects to this `dispatch` method. It then gets spread to the reducers in our application.

In the typical React application we usually don't use `getState` and `subscribe` directly because there is a helper (we will see it in the next sections) that wires our components with the store and effectively `subscribes` for changes. As part of this subscription we also receive the current state so we don't have to call `getState` ourself. `replaceReducer` is kind of an advanced API and it swaps the reducer currently used by the store. I personally never used this method.

Reducer

The reducer function is probably the most *beautiful* part of Redux. There are two characteristics of the reducer that are quite important and without them we basically have a broken pattern.

- It must be a pure function - it means that the function should return the exact same output every time, given the same input.
- It should have no side effects - stuff like accessing a global variable, making an async call or waiting for a promise to resolve have no place in here.

Here is a simple counter reducer:

```
const counterReducer = function (state, action) {
  if (action.type === ADD) {
    return { value: state.value + 1 };
  } else if (action.type === SUBTRACT) {
    return { value: state.value - 1 };
  }
  return { value: 0 };
};
```

There are no side effects and we return a brand new object every time. We accumulate the new value based on the previous state and the incoming action type.

In the next lesson, we will see how the react-redux module can be used to connect Redux to React components

Wiring to React components

This lesson introduces the react-redux module and how it is used to connect Redux with React components

WE'LL COVER THE FOLLOWING

- Wiring to React components

Wiring to React components

If we talk about Redux in the context of React we almost always mean react-redux module. It provides two things that help connecting Redux to our components.

- `<Provider>` component - it's a component that accepts our store and makes it available for the children down the React tree via the React's context API. For example:

```
<Provider store={ myStore }>
  <MyApp />
</Provider>
```

We usually have a single place in our app where we use it.

- `connect` function - it is a function that does the subscribing for updates in the store and re-renders our component. It implements a higher-order component. Here is its signature:

```
connect(
  [mapStateToProps],
  [mapDispatchToProps],
  [mergeProps],
  [options]
)
```

`mapStateToProps` parameter is a function that accepts the current state and

must return a set of key-value pairs (an object) that are being sent as props to our React component. For example:

```
const mapStateToProps = state => ({
  visible: state.visible
});
```

`mapDispatchToProps` is a similar one but instead of the `state`, it receives a `dispatch` function. Here is the place where we can define a prop for dispatching actions.

```
const mapDispatchToProps = dispatch => ({
  changeVisibility: value => dispatch(changeVisibility(value))
});
```

`mergeProps` combines both `mapStateToProps` and `mapDispatchToProps`. The props sent to the component gives us the opportunity to accumulate better props. Like for example if we need to fire two actions we may combine them to a single prop and send that to React. `options` accepts couple of settings that control how the connection works. In the following section, we will look into creating an app using Redux.

Let's use all we have learned about Redux to make a simple counter app in the next lesson

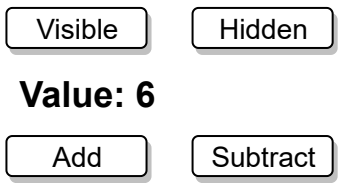
Simple Counter App

This lesson uses all we have learned about Redux in the previous few lessons to make a Simple Counter Application. Let's dive in!

WE'LL COVER THE FOLLOWING

- Modeling the actions
- Store and its reducers
- Selectors
- React components

Let’s create a simple counter app that uses all the APIs above.



Visual: Simple Counter App

The “Add” and “Subtract” buttons will change a value in our store. “Visible” and “Hidden” will control its visibility.

Modeling the actions

For me, every Redux feature starts with modeling the action types and defining what state we want to keep. In our case, we have three operations going on - adding, subtracting and managing visibility. So we will go with the following:

```
const ADD = 'ADD';
const SUBTRACT = 'SUBTRACT';
const CHANGE_VISIBILITY = 'CHANGE_VISIBILITY';

const add = () => ({ type: ADD });
const subtract = () => ({ type: SUBTRACT });
```

```
const changeVisibility = visible => ({
  type: CHANGE_VISIBILITY,


  visible
});
```

Store and its reducers

There is something that we didn't talk about while explaining the store and reducers. We usually have more than one reducer because we want to manage multiple things. The store is one though and we in theory have only one state object. What happens in most of the apps running in production is that the application state is a composition of slices. Every slice represents a part of our system. In this very small example we have counting and visibility slices. So our initial state looks like this:

```
const initialState = {
  counter: {
    value: 0
  },
  visible: true
};
```

We must define separate reducers for both parts. This is to introduce some flexibility and to improve the readability of our code. Imagine if we have a giant app with ten or more state slices and we keep working within a single function. It will be too difficult to manage.

Redux comes with a helper that allows us to target a specific part of the state and assign a reducer to it. It is called `combineReducers`:

```
import { createStore, combineReducers } from 'redux';

const rootReducer = combineReducers({
  counter: function A() { ... },
  visible: function B() { ... }
});
const store = createStore(rootReducer);
```

Function `A` receives only the `counter` slice as a state and needs to return only that part. The same goes for `B`, it also receives only the `counter` slice as a state. Accepts a boolean (the value of `visible`) and must return a boolean.

The reducer for our counter slice should take into account both actions `ADD,`

`SUBTRACT` and based on them calculate the new `counter` state.

```
const counterReducer = function (state, action) {
  if (action.type === ADD) {
    return { value: state.value + 1 };
  } else if (action.type === SUBTRACT) {
    return { value: state.value - 1 };
  }
  return state || { value: 0 };
};
```

Every reducer is fired at least once when the store is initialized. In that very first run the `state` is `undefined` and the `action` is `{ type: "@@redux/INIT"}`. In this case our reducer should return the initial value of our data - `{ value: 0 }`.

The reducer for the visibility is pretty similar except that it waits for `CHANGE_VISIBILITY` action:

```
const visibilityReducer = function (state, action) {
  if (action.type === CHANGE_VISIBILITY) {
    return action.visible;
  }
  return true;
};
```

And at the end we have to pass both reducers to `combineReducers` so we create our `rootReducer`.

```
const rootReducer = combineReducers({
  counter: counterReducer,
  visible: visibilityReducer
});
```

Selectors

Before moving to the React components we have to mention the concept of a *selector*. From the previous section we know that our state is usually divided into different parts. We have dedicated reducers to update the data but when it comes to fetching it we still have a single object. Here is the place where the selectors come in handy. The selector is a function that accepts the whole state object and extracts only the information that we need. For example in our small app we need two of those:

```
const getCounterValue = state => state.counter.value;
const getVisibility = state => state.visible;
```

A counter app is too small to see the real power of writing such helpers. However, for a big project, it is quite different. And it is not just about saving a few lines of code. Neither is it about readability. Selectors come with this stuff but they are also contextual and may contain logic. Since they have access to the whole state they are able to answer business logic related questions. Like for example “Is the user authorized to do X while being on page Y”. This may be done in a single selector.

React components

Let’s first deal with the UI that manages the visibility of the counter.

```
function Visibility({ changeVisibility }) {
  return (
    <div>
      <button onClick={ () => changeVisibility(true) }>
        Visible
      </button>
      <button onClick={ () => changeVisibility(false) }>
        Hidden
      </button>
    </div>
  );
}

const VisibilityConnected = connect(
  null,
  dispatch => ({
    changeVisibility: value => dispatch(changeVisibility(value))
  })
)(Visibility);
```

We have two buttons `Visible` and `Hidden`. They both fire `CHANGE_VISIBILITY` action but the first one passes `true` as a value while the second one `false`. The `VisibilityConnected` component class gets created as a result of the wiring done via Redux’s `connect`. Notice that we pass `null` as `mapStateToProps` because we don’t need any of the data in the store. We just need to `dispatch` an action.

The second component is slightly more complicated. It is named `Counter` and renders two buttons and the counter value.

```
function Counter({ value, add, subtract }) {
```

```
  return (
    <div>
      <p>Value: { value }</p>

      <button onClick={ add }>Add</button>
      <button onClick={ subtract }>Subtract</button>
    </div>
  );
}

const CounterConnected = connect(
  state => ({
    value: getCounterValue(state)
  }),
  dispatch => ({
    add: () => dispatch(add()),
    subtract: () => dispatch(subtract())
  })
)(Counter);
```

We now need both `mapStateToProps` and `mapDispatchToProps` because we want to read data from the store and dispatch actions. Our component receives three props - `value`, `add` and `subtract`.

The very last bit is an `App` component where we compose the application.

```
function App({ visible }) {
  return (
    <div>
      <VisibilityConnected />
      { visible && <CounterConnected /> }
    </div>
  );
}
const AppConnected = connect(
  state => ({
    visible: getVisibility(state)
  })
)(App);
```

We again need to `connect` our component because we want to control the visibility of the counter. The `getVisibility` selector returns a boolean that indicates whether `CounterConnected` will be rendered or not.

In the next and final lesson of this chapter, we will present some final thoughts and executable code of Simple Counter App

Final Thoughts and Code for Counter App

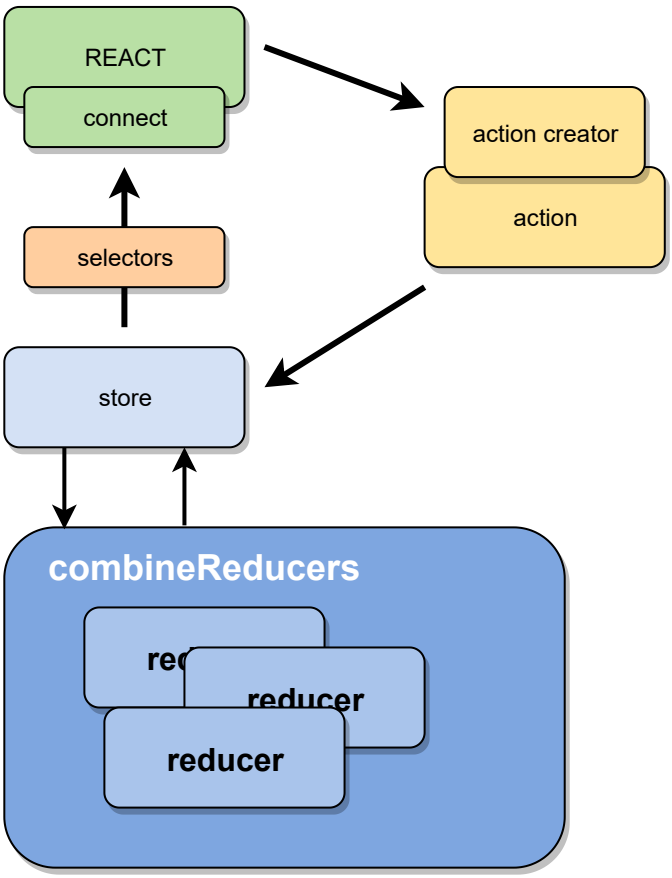
This lesson presents some final thoughts and complete, executable code for Simple Counter Application.

WE'LL COVER THE FOLLOWING

- Final thoughts
- Quick Quiz on Redux

Final thoughts

Redux is a wonderful pattern. Over the years the JavaScript community developed the idea and enhanced it with a couple of new terms. A typical redux application may look like this:



Redux Architecture

We can conclude that Redux itself is a pretty simple pattern. It teaches very

useful techniques but unfortunately, it is very often not enough. Sooner or

later we have to introduce more concepts/patterns. Which of course is not that bad. We just have to plan for it.

```
import React from 'react';
import ReactDOM from 'react-dom';
import './style.css';



import { Provider, connect } from 'react-redux';
import { createStore, combineReducers } from 'redux';

// Action creators

const ADD = 'ADD';
const SUBTRACT = 'SUBTRACT';
const CHANGE_VISIBILITY = 'CHANGE_VISIBILITY';

const add = () => ({ type: ADD });
const subtract = () => ({ type: SUBTRACT });
const changeVisibility = visible => ({ type: CHANGE_VISIBILITY, visible });

// Reducers

const initialState = {
  counter: {
    value: 0
  },
  visible: true
};
const counterReducer = function (state, action) {
  console.log(state, action);
  if (action.type === ADD) {
    return { value: state.value + 1 };
  } else if (action.type === SUBTRACT) {
    return { value: state.value - 1 };
  }
  return state || { value: 0 };
};
const visibilityReducer = function (state, action) {
  if (action.type === CHANGE_VISIBILITY) {
    return action.visible;
  }
  return true;
};
const rootReducer = combineReducers({
  counter: counterReducer,
  visible: visibilityReducer
});

// Selectors

const getCounterValue = state => state.counter.value;
const getVisibility = state => state.visible;

// Store creation
```

```
const store = createStore(rootReducer, initialState);


// React components

function Counter({ value, add, subtract }) {
  return (
    <div>
      <p>Value: { value }</p>
      <button onClick={ add }>Add</button>
      <button onClick={ subtract }>Subtract</button>
    </div>
  );
}
const CounterConnected = connect(
  state => ({
    value: getCounterValue(state)
  }),
  dispatch => ({
    add: () => dispatch(add()),
    subtract: () => dispatch(subtract())
  })
)(Counter);

function Visibility({ changeVisibility }) {
  return (
    <div>
      <button onClick={ () => changeVisibility(true) }>Visible</button>
      <button onClick={ () => changeVisibility(false) }>Hidden</button>
    </div>
  );
}
const VisibilityConnected = connect(
  null,
  dispatch => ({
    changeVisibility: value => dispatch(changeVisibility(value))
  })
)(Visibility);



function App({ visible }) {
  return (
    <div>
      <VisibilityConnected />
      { visible && <CounterConnected /> }
    </div>
  );
}

const AppConnected = connect(
  state => ({
    visible: getVisibility(state)
  })
)(App);

ReactDOM.render(<Provider store={ store }><AppConnected /></Provider>, document.getElementByI
```

Quick Quiz on Redux

1 There is only one _____ in Redux.

COMPLETED 0%

1 of 3

This is about it on Data Flow in React. There are a few other important topics that we are going to discuss in the next chapter.

Dependency Injection - Using React's context (prior v. 16.3)

In this lesson, we will be learning to use dependency injection as a means to manage dependencies in React.

WE'LL COVER THE FOLLOWING

- Dependency Injection
- Using React's context (prior v. 16.3)

Dependency Injection

Many of the modules/components that we write have dependencies. Proper management of these dependencies is critical for the success of the project. There is a technique (most people consider it a *pattern*) called *dependency injection* that helps to solve the problem.

In React the need for dependency injector is easily visible. Let's consider the following application tree:

```
// Title.jsx
export default function Title(props) {
  return <h1>{ props.title }</h1>;
}

// Header.jsx
import Title from './Title.jsx';

export default function Header() {
  return (
    <header>
      <Title />
    </header>
  );
}

// App.jsx
import Header from './Header.jsx';

class App extends React.Component {
  constructor(props) {
```

```
  constructor(props) {
    super(props);
    this.state = { title: 'React in patterns' };
  }
  render() {
    return <Header />;
  }
};
```

The string “React in patterns” should somehow reach the `Title` component. The direct way of doing this is to pass it from `App` to `Header` and then `Header` pass it down to `Title`. However, this may work for these three components but what happens if there are multiple properties and deeper nesting. Lots of components will act as a proxy, passing properties to their children.

We already saw how the higher-order component may be used to inject data. Let’s use the same technique to inject the `title` variable:

```
// inject.jsx
const title = 'React in patterns';

export default function inject(Component) {
  return class Injector extends React.Component {
    render() {
      return (
        <Component
          {...this.props}
          title={ title }
        />
      )
    }
  };
}

// -----------------------------------
// Header.jsx
import inject from './inject.jsx';
import Title from './Title.jsx';

var EnhancedTitle = inject(Title);
export default function Header() {
  return (
    <header>
      <EnhancedTitle />
    </header>
  );
}
```

The `title` is hidden in a middle layer (higher-order component) where we pass it as a prop to the original `Title` component. That’s all nice but it solves only half of the problem. Now we don’t have to pass the `title` down the tree but how this data reaches the `inject.jsx` helper.

Using React's context (prior v. 16.3)

In v16.3 React's team introduced a new version of the context API and if you are going to use that version or above you'd probably skip this section.

React has the concept of *context*. The *context* is something that every React component has access to. It's like an event bus but for data. A single *store* which we access from everywhere.

```
// a place where we will define the context
var context = { title: 'React in patterns' };

class App extends React.Component {
  getChildContext() {
    return context;
  }
  ...
};
App.childContextTypes = {
  title: React.PropTypes.string
};

// a place where we use the context
class Inject extends React.Component {
  render() {
    var title = this.context.title;
    ...
  }
}
Inject.contextTypes = {
  title: React.PropTypes.string
};
```

Notice that we have to specify the exact signature of the context object with `childContextTypes` and `contextTypes`. If those are not specified then the `context` object will be empty. That can be a little bit frustrating because we may have lots of stuff to put there. That is why it is a good practice that our `context` is not just a plain object but it has an interface that allows us to store and retrieve data. For example:

```
// dependencies.js
export default {
  data: {},
  get(key) {
    return this.data[key];
  },
  register(key, value) {
    this.data[key] = value;
  }
}
```

Then, if we go back to our example, the `App` component may look like this:

```
import dependencies from './dependencies';

dependencies.register('title', 'React in patterns');

class App extends React.Component {
  getChildContext() {
    return dependencies;
  }
  render() {
    return <Header />;
  }
};
App.childContextTypes = {
  data: React.PropTypes.object,
  get: React.PropTypes.func,
  register: React.PropTypes.func
};
```

And our `Title` component gets it's data through the context:

```
// Title.jsx
export default class Title extends React.Component {
  render() {
    return <h1>{ this.context.get('title') }</h1>
  }
}
Title.contextTypes = {
  data: React.PropTypes.object,
  get: React.PropTypes.func,
  register: React.PropTypes.func
};
```

Ideally we don't want to specify the `contextTypes` every time we need access to the context. This detail may be wrapped again in a higher-order component. And even better, we may write a utility function that is more descriptive and helps us declare the exact wiring. I.e instead of accessing the context directly with `this.context.get('title')` we ask the higher-order component to get what we need and pass it as props to our component. For example:

```
// Title.jsx
import wire from './wire';

function Title(props) {
  return <h1>{ props.title }</h1>;
```

```
}

export default wire(Title, ['title'], function resolve(title) {

  return { title };
});
```

The `wire` function accepts a React component, then an array with all the needed dependencies (which are `register`ed already) and then a function which you may call `mapper`. It receives what is stored in the context as a raw data and returns an object which is later used as props for our component (`Title`). In this example we just pass what we get - a `title` string variable. However, in a real app, this could be a collection of data stores, configuration or something else.

Here is what the `wire` function looks like:

```
export default function wire(Component, dependencies, mapper) {
  class Inject extends React.Component {
    render() {
      var resolved = dependencies.map(
        this.context.get.bind(this.context)
      );
      var props = mapper(...resolved);

      return React.createElement(Component, props);
    }
  }
  Inject.contextTypes = {
    data: React.PropTypes.object,
    get: React.PropTypes.func,
    register: React.PropTypes.func
  };
  return Inject;
};
```

`Inject` is a higher-order component that gets access to the context and retrieves all the items listed under `dependencies` array. The `mapper` is a function that receives the `context` data and transforms it into props for our component.

Below is the full code that injects dependencies into the Title component through context:

```
export default {
  name: 'React in patterns'
};
```

In the next section, we will discuss how dependency injections work in later versions of React.

Dependency Injection - Using React's context (v. 16.3 and above)

This lesson discusses how the context API has improved the implementation of dependency injection in later versions.

WE'LL COVER THE FOLLOWING

- Using the module system
- Final thoughts

For years the context API was not really recommended by Facebook. They mentioned in the official docs that the API is not stable and may change. And that is exactly what happened. In version 16.3, we got a new one which is more natural and easy to work with.

Let's use the same example with the string that needs to reach a `<Title>` component.

We will start by defining a file that will contain our context initialization:

```
// context.js
import { createContext } from 'react';

const Context = createContext({});

export const Provider = Context.Provider;
export const Consumer = Context.Consumer;
```

`createContext` returns an object that has `.Provider` and `.Consumer` properties. Those are actually valid React classes. The `Provider` accepts our context in the form of a `value` prop. The consumer is used to access the context and basically read data from it. And because they usually live in different files it is a good idea to create a single place for their initialization.

Let's say that our `App` component is the root of our tree. That is where we

have to pass the context.

```
import { Provider } from './context';

const context = { title: 'React In Patterns' };

class App extends React.Component {
  render() {
    return (
      <Provider value={ context }>
        <Header />
      </Provider>
    );
  }
};
```

The wrapped components and their children now share the same context. The `<Title>` component is the one that needs the `title` string so that is the place where we use the `<Consumer>`.

```
import { Consumer } from './context';

function Title() {
  return (
    <Consumer>{
      ({ title }) => <h1>Title: { title }</h1>
    }</Consumer>
  );
}
```

Notice that the `Consumer` class uses the function as children (render prop) pattern to deliver the context.

The new API feels easier to understand and eliminates the boilerplate. It is still pretty new but looks promising. It opens a whole new range of possibilities.

Using the module system

If we don't want to use the context there are a couple of other ways to achieve the injection. They are not exactly React specific, but worth mentioning nevertheless. One of them is to use the module system.

As we know, the typical module system in JavaScript has a caching mechanism. It's nicely noted in Node's documentation:

> Modules are cached after the first time they are loaded. This means (among other things) that every call to require('foo') will get exactly the same object returned, if it would resolve to the same file.

> Multiple calls to require('foo') may not cause the module code to be executed multiple times. This is an important feature. With it, "partially done" objects can be returned, thus allowing transitive dependencies to be loaded even when they would cause cycles.

How does that help in our injection? Well, if we export an object we are actually exporting a singleton and every other module that imports the file will get the same object. This allows us to `register` our dependencies and later `fetch` them in another file.

Let's create a new file called `di.jsx` with the following content:

```
var dependencies = {};

export function register(key, dependency) {
  dependencies[key] = dependency;
}

export function fetch(key) {
  if (dependencies[key]) return dependencies[key];
  throw new Error(`"${ key } is not registered as dependency.`);
}

export function wire(Component, deps, mapper) {
  return class Injector extends React.Component {
    constructor(props) {
      super(props);
      this._resolvedDependencies = mapper(...deps.map(fetch));
    }
    render() {
      return (
        <Component
          {...this.state}
          {...this.props}
          {...this._resolvedDependencies}
        />
      );
    }
  };
}
```

We'll store the dependencies in `dependencies` global variable (it's global for our module, not for the whole application). We then export two functions

`register` and `fetch` that write and read entries. It looks a little bit like implementing setter and getter functions against a simple JavaScript object.

Then we have the `wire` function that accepts our React component and returns a higher-order component. In the constructor of that component we are resolving the dependencies and later, while rendering the original component, we pass them as props. We follow the same pattern where we describe what we need (`deps` argument) and extract the needed props with a `mapper` function.

Having the `di.jsx` helper, we are once again able to register our dependencies at the entry point of our application (`app.jsx`) and inject them wherever (`Title.jsx`) we need.

```
// app.jsx
import Header from './Header.jsx';
import { register } from './di.jsx';

register('my-awesome-title', 'React in patterns');

class App extends React.Component {
  render() {
    return <Header />;
  }
};

// -----------------------------------
// Header.jsx
import Title from './Title.jsx';

export default function Header() {
  return (
    <header>
      <Title />
    </header>
  );
}

// -----------------------------------
// Title.jsx
import { wire } from './di.jsx';

var Title = function(props) {
  return <h1>{ props.title }</h1>;
};

export default wire(
  Title,
  ['my-awesome-title'],
  title => ({ title })
);
```

If we look at the `Title.jsx` file we'll see that the actual component and the

wiring may live in different files. That way the component and the mapper function become easily unit testable.

Final thoughts

Dependency injection is a tough problem. Especially in JavaScript. However, employing proper dependency management is a key process of every development cycle. The JavaScript ecosystem offers different tools that can be used for dependency management and we as developers should pick the ones that fit our needs best. Following is the complete dependency injection code for the example we have been following in this lesson:

```
import React from 'react';
import PropTypes from 'prop-types';
import { wire } from './di.jsx';

var Title = function (props) {
  return <h1>{ props.title }</h1>;
};

Title.propTypes = {
  title: PropTypes.string
};

export default wire(Title, ['my-awesome-title'], title => ({ title }));
```

Now that we know how to handle dependencies in React, the next step is to look at how to present elements in React. In the following chapter, we will discuss some miscellaneous topics including styling and integration of third-party libraries.

Styling React Components

This lesson introduces some common methods used to style React components.

WE'LL COVER THE FOLLOWING

- The good old CSS class
- Inline styling
- CSS modules
- Styled-components
- Final thoughts

React is a view layer. It kind of controls the markup rendered in the browser. And we know that the styling with CSS is tightly connected to the markup on the page. There are couple of approaches for styling React applications and in this section we will go through the most popular ones.

The good old CSS class

JSX syntax is pretty close to HTML syntax. As such we have almost the same tag attributes and we may still style using CSS classes. Classes which are defined in an external `.css` file. The only caveat is using `className` and not `class`. For example:

```
<h1 className='title'>Styling</h1>
```

Inline styling

The inline styling works just fine. Similarly to HTML we are free to pass styles directly via a `style` attribute. However, while in HTML the value is a string, in JSX it must be an object.

```
const inlineStyles = {
```

```
  color: 'red',
  fontSize: '10px',
  marginTop: '2em',

  'border-top': 'solid 1px #000'
};

<h2 style={ inlineStyles }>Inline styling</h2>
```

Since we write the styles in JavaScript we have some limitations from a syntax point of view. If we want to keep the original CSS property names we have to put them in quotes. If not then we have to follow the camel case convention. However, writing styles in JavaScript is quite interesting and may be a lot more flexible than the plain CSS. Like for example inheriting of styles:

```
const theme = {
  fontFamily: 'Georgia',
  color: 'blue'
};
const paragraphText = {
  ...theme,
  fontSize: '20px'
};
```

We have some basic styles in `theme` and we mix them with what is in `paragraphText`. Shortly, we are able to use the whole power of JavaScript to organize our CSS. What matters at the end is that we generate an object that goes to the `style` attribute.

CSS modules

CSS modules is building on top of what we said so far. If we don't like the JavaScript syntax then we may use CSS modules and we will be able to write plain CSS. Usually this library plays its role at bundling time. It is possible to hook it as part of the transpilation step but normally is distributed as a build system plugin.

Here is a quick example to get an idea how it works:

```
/* style.css */
.title {
  color: green;
}

// App.jsx
import styles from "./style.css";
```

```
function App() {
  return <h1 style={ styles.title }>Hello world</h1>;
}
```

That is not possible by default but with CSS modules we may import directly a plain CSS file and use the classes inside.

And when we say *plain CSS* we don't mean that it is exactly like the normal CSS. It supports some really helpful composition techniques. For example:

```
.title {
  composes: mainColor from "./brand-colors.css";
}
```

Styled-components

Styled-components took another direction. Instead of inlining styles the library provides a React component. We then use this component to represent a specific look and feel. For example, we may create a `Link` component that has certain styling and use that instead of the `<a>` tag.

```
const Link = styled.a`
  text-decoration: none;
  padding: 4px;
  border: solid 1px #999;
  color: black;
`;

<Link href='http://google.com'>Google</Link>
```

There is again a mechanism for extending classes. We may still use the `Link` component but change the text color like so:

```
const AnotherLink = styled(Link)`
  color: blue;
`;

<AnotherLink href='http://facebook.com'>Facebook</AnotherLink>
```

For me styled-components are probably by far the most interesting approach for styling in React. It is quite easy to create components for everything and forget about the styling. If your company has the capacity to create a design system and build a product with it then this option is probably the most suitable one.

Final thoughts

There are multiple ways to style your React application, there is no right or wrong way. As is the case with most things in JavaScript today, you have to pick the one that fits better in your context.

Here is the final code for all the different methods we learned to style our components in:

```
import React from 'react';
import ReactDOM from 'react-dom';

const inlineStyles = {
  color: 'red',
  fontSize: '10px',
  marginTop: '2em',
  'border-top': 'solid 1px #000'
};

const theme = {
  fontFamily: 'Georgia',
  color: 'blue'
};
const paragraphText = {
  ...theme,
  fontSize: '20px'
};

function App() {
  return (
    <section>
      <h1 className='title'>Styling</h1>
      <h2 style={inlineStyles}>Inline styling</h2>
      <p style={paragraphText}>This is a paragraph here!</p>
    </section>
  );
}

const rootElement = document.getElementById("root");
ReactDOM.render(<App />, rootElement);
```

Now that we have learned multiple ways to style React components, in the next lesson we will move forward to look into integrating third-party libraries within our applications.

Integration Of Third-Party Libraries

In this lesson, we will learn how to safely integrate third-party components in React apps, when external services are required.

WE'LL COVER THE FOLLOWING

- The example
- Force a single-render
- Initializing the plugin
- Controlling the plugin using React
- Final thoughts

React is probably one of the best choices for building UI because of its good design, support, and community. However, there are cases where we want to use an external service or we want to integrate something completely different. We all know that React works heavily with the actual DOM and basically controls what's rendered on the screen. That's why integrating third-party components may be tricky. In this section we will see how to mix React and jQuery's UI plugin and do it safely.

The example

Let's take tag-it jQuery plugin as an example. It transforms an unordered list to input field for managing tags:

```
<ul>
  <li>JavaScript</li>
  <li>CSS</li>
</ul>
```

to:

To make it work we have to include jQuery, jQuery UI and the *tag-it* plugin code. It works like this:

```
$('<dom element selector>').tagit();
```

We select a DOM element and call `tagit()`.

Now, let's create a simple React app that will use the plugin:

```
// Tags.jsx
class Tags extends React.Component {
  componentDidMount() {
    // initialize tagit
    $(this.refs.list).tagit();
  }
  render() {
    return (
      <ul ref="list">
      {
        this.props.tags.map(
          (tag, i) => <li key={ i }>{ tag } </li>
        )
      }
      </ul>
    );
  }
};

// App.jsx
class App extends React.Component {
  constructor(props) {
    super(props);

    this.state = { tags: ['JavaScript', 'CSS' ] };
  }
  render() {
    return (
      <div>
        <Tags tags={ this.state.tags } />
      </div>
    );
  }
}

ReactDOM.render(<App />, document.querySelector('#container'));
```

The entry point is our `App` class. It uses the `Tags` component that displays an unordered list based on the passed `tags` prop. When React renders the list on

the screen we know that we have a `<ul>` tag so we can hook it up to the jQuery plugin.

Force a single-render

The very first thing that we have to do is to force a single-render of the `Tags` component. That is because when React adds the elements in the actual DOM we want to pass the control of them to jQuery. If we skip this both React and jQuery will work on the same DOM elements without knowing of each other. To achieve a single-render we have to use the lifecycle method `shouldComponentUpdate` like so:

```
class Tags extends React.Component {
  shouldComponentUpdate() {
    return false;
  }
  ...
```

By always returning `false` here we are saying that our component will never re-render. If defined `shouldComponentUpdate` is used by React to understand whether to trigger `render` or not. That is ideal for our case because we want to place the markup on the page using React but we don't want to rely on it after that.

Initializing the plugin

React gives us an API for accessing actual DOM nodes. We have to use the `ref` attribute on a node and later reach that node via `this.refs.componentDidMount` which is a proper lifecycle method for initializing the *tag-it* plugin. That's because we get it called when React mounts the result of the `render` method.

```
class Tags extends React.Component {
  ...
  componentDidMount() {
    this.list = $(this.refs.list);
    this.list.tagit();
  }
  render() {
    return (
      <ul ref='list'>
      {
        this.props.tags.map(
          (tag, i) => <li key={ i }>{ tag } </li>
        )
      }
      </ul>
    );
```

```
  );
  }
  ...
```

The code above together with `shouldComponentUpdate` leads to React rendering the `<ul>` with two items and then *tag-it* transforms it to a working tag editing widget.

Controlling the plugin using React

Let's say that we want to programmatically add a new tag to the already running *tag-it* field. Such action will be triggered by the React component and needs to use the jQuery API. We have to find a way to communicate data to `Tags` component but still keep the single-render approach.

To illustrate the whole process we will add an input field to the `App` class and a button which if clicked will pass a string to `Tags` component.

```
class App extends React.Component {
  constructor(props) {
    super(props);

    this._addNewTag = this._addNewTag.bind(this);
    this.state = {
      tags: ['JavaScript', 'CSS' ],
      newTag: null
    };
  }
  _addNewTag() {
    this.setState({ newTag: this.refs.field.value });
  }
  render() {
    return (
      <div>
        <p>Add new tag:</p>
        <div>
          <input type='text' ref='field' />
          <button onClick={ this._addNewTag }>Add</button>
        </div>
        <Tags
          tags={ this.state.tags }
          newTag={ this.state.newTag } />
      </div>
    );
  }
}
```

We use the internal state as a data storage for the value of the newly added field. Every time we click the button we update the state and trigger re-rendering of `Tags` component. However, because of `shouldComponentUpdate` we have no any updates on the screen. The only one change is that we get a

new value of the `newTag` prop which may be captured via another lifecycle method - `componentWillReceiveProps`.

The full code is as follows:

```
class Tags extends React.Component {
  componentDidMount() {
    this.list = $(this.refs.list);
    this.list.tagit();
  }
  shouldComponentUpdate() {
    return false;
  }
  componentWillReceiveProps(newProps) {
    this.list.tagit('createTag', newProps.newTag);
  }
  render() {
    return (
      <ul ref='list'>
      {
        this.props.tags.map(
          (tag, i) => <li key={ i }>{ tag } </li>
        )
      }
      </ul>
    );
  }
};
```

Final thoughts

Even though React is manipulating the DOM tree we are able to integrate third-party libraries and services. The available lifecycle methods give us enough control on the rendering process so they are the perfect bridge between React and non-React code.

React And Separation Of Concerns

In this lesson, we will learn how React and its ecosystem have a separation of concerns despite markup, styles, and logic living in the same JavaScript land.

WE'LL COVER THE FOLLOWING

- Styling
- Logic
- Markup
- Conclusion

Years ago when Facebook announced their JSX syntax we had a wave of comments how this was against some of the well established good practices. The main point of most people was that it violates the separation of concerns. They said that React and its JSX are mixing HTML, CSS and JavaScript which were suppose to be separated.

In this article we will see how React and its ecosystem has quite good separation of concerns. We will prove that markup, styles and logic may live in the same JavaScript land and still be separated.

Styling

React components render to DOM elements. Nothing stops us to use the good old `class` attribute and attach a CSS class to the produced HTML element. The only difference is that the attribute is called `className` instead of `class`. The rest still works which means that if we want we may put our styles into external `.css` files. Following this approach we are not breaking the separation of concerns principle and still build a React app using JSX.

```
// assets/css/styles.css
.pageTitle {
  color: pink;
}
```

```
}

// assets/js/app.js
function PageTitle({ text }) {
  return <h1 className='pageTitle'>{ text }</h1>;
}
```

The “problem” became a problem when developers started talking about “CSS in JavaScript”. Back in 2014 this looked weird and wrong. However, the next couple of years showed that this is not that bad. Let’s take the following example:

```
function UserCard({ name, avatar }) {
  const cardStyles = {
    padding: '1em',
    boxShadow: '0px 0px 45px 0px #000'
  };
  const avatarStyles = {
    float: 'left',
    display: 'block',
    marginRight: '1em'
  };
  return (
    <div style={cardStyles}>
      <img src={avatar} width="50" style={avatarStyles} />
      <p>{name}</p>
    </div>
  );
}
```

The pain point is where we are mixing CSS and markup or we may say mixing styling and structure. To solve the issue let’s keep `UserCard` still responsible for the structure, but extract the styling out into dedicated components `Card` and `Avatar`:

```
function Card({ children }) {
  const styles = {
    padding: '1em',
    boxShadow: '0px 0px 45px 0px #000',
    maxWidth: '200px'
  };
  return <div style={styles}>{children}</div>;
}
function Avatar({ url }) {
  const styles = {
    float: 'left',
    display: 'block',
    marginRight: '1em'
  };
  return <img src={url} width="50" style={styles} />;
}
```

Then `UserCard` component becomes simpler and has no styling concerns:

```
function UserCard({ name, avatar }) {
  return (
    <Card>
      <Avatar url={avatar} />
      <p>{name}</p>
    </Card>
  );
}
```

So, as we can see, it is all a matter of composition. React even makes our applications more compact because everything is defined as a reusable component and lives in the same context - JavaScript.

There are bunch of libraries that help writing maintainable CSS in JavaScript (and to be more specific in React ecosystem) like Glamorous and styled-components. The result of such libraries is usually ready for use component that encapsulates the styling and renders a specific HTML tag.

Logic

Very often we write logic inside our React components which is more then clicking a button and showing a message. The snippet below demonstrates a component that fetches data from a fake API and renders users on the screen.

```
class App extends React.Component {
  constructor(props) {
    super(props);

    this.state = {
      loading: false,
      users: null,
      error: null
    };
  }
  componentDidMount() {
    this.setState({ loading: true }, () => {
      fetch('https://jsonplaceholder.typicode.com/users')
        .then(response => response.json())
        .then(users => this.setState({ users, loading: false }))
        .catch(error => this.setState({ error, loading: false }));
    });
  }
  render() {
    const { loading, users, error } = this.state;

    if (isRequestInProgress) return <p>Loading</p>;
    if (error) return <p>Ops, sorry. No data loaded.</p>;
    if (users) return users.map(({ name }) => <p>{name}</p>);
```

```
    return null;
  }
}
```

Quite a lot of things are happening. When first rendered the component shows nothing - `null`. Then we get the life-cycle `componentDidMount` method fired where we set the `loading` flag to `true` and fire the API request. While the request is in flight we display a paragraph containing the text `"Loading"`. In the end, if everything is ok we turn `loading` to false and render list of user names. In case of error we display `"Ops, sorry. No data loaded"`.

Now you can say that the `App` component is kind of violating the separation of concerns. It contains data fetching and data representation. There are couple of ways to solve this problem one of which is FaCC (Function as Child Components). Let's write a `FetchUsers` component that will take care for the API request.

```
class FetchUsers extends React.Component {
  constructor(props) {
    super(props);

    this.state = {
      loading: false,
      users: null,
      error: null
    };
  }
  componentDidMount() {
    this.setState({ loading: true }, () => {
      fetch('https://jsonplaceholder.typicode.com/users')
        .then(response => response.json())
        .then(users => this.setState({ users, loading: false }))
        .catch(error => this.setState({ error, loading: false }));
    });
  }
  render() {
    const { loading, users, error } = this.state;

    return this.props.children({ loading, users, error });
  }
}
```

The very first thing that we notice is that the constructor and `componentDidMount` method are just copy-pasted from the `App` component. The difference is that we render nothing (no data representation) but call the `children` as a function passing the status/result of the request. Having `FetchUsers` we may transform our `App` into a stateless component:

```
function App() {
```

```
function App() {
  return (
    <FetchUsers>
      {({ loading, users, error }) => {
        if (loading) return <p>Loading</p>;
        if (error) return <p>Ops, sorry. No data loaded.</p>;
        if (users) return users.map(({ name }) => <p>{name}</p>);
        return null;
      }}
    </FetchUsers>
  );
}
```

At this point, our markup is separated from the logic. We still operate with the same data and as a bonus, we have this reusable `FetchUsers` component that may be dropped anywhere.

Markup

JSX syntax is following the XML/HTML semantics and as such comes with a huge benefit - composability. You can say that React is one level up over the HTML because it allows us to group complex markup into a single component. For example we have a `<header>` with some `<h1>`, `<nav>` and `<p>` tags inside. We may easily create a `<Header>` component and put all those bits inside. We still keep them together but now they are easy to move around. Perhaps, there are places where we write some logic directly into the markup like so:

```
function CallToActionButton({ service, token }) {
  return <button onClick={ () => service.request(token) } />;
}

<CallToAction server={ service } token={ token } />
```

In such cases, it is recommended to use composition and remove any app logic concerns out of the presentation.

```
function CallToActionButton({ onButtonClicked }) {
  return <button onClick={ onButtonClicked } />;
}

<CallToAction server={ () => service.request(token) } />
```

It is just a matter of spotting those bits and changing them on time.

Conclusion

React is not against the separation of concerns. It is all about design and composition. There are patterns that help us to compose and logically separate our apps. We can still write well-organized programs with clearly defined responsibilities.

Where to from here

Wondering where to go next? Our website is chalk full of other React courses that you can take!

We're glad you've gotten to the end of this course. Thank you for taking the time out and reading it. We hope you've gained some invaluable knowledge about React and feel more confident as a React developer. If you want to further level up your React skills, there are a number of courses on our website that you can consider taking.

Course	Author	What its about
Practical Redux	**Mark Erikson**	how to use Redux in real-world apps
Understanding Redux: A Beginner's Guide To State Management	**Ohans Emmanuel**	a gateway into the mechanics of using Redux as an application building platform.
Building Tesla's Battery Range Calculator with React & Redux	**Matthew Choi**	This course will help you to advance your React and Redux skills by building a battery range calculator

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And that's a wrap! If you have any feedback, suggestions, or comments, feel free to drop us an email or a comment on our forum!